# A Little R Survival Kit

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# Chapter 1

# A Little R Survival Kit: Essential Data Analysis using R

# Preface

In the 21st century, nobody does statistics by pen and paper. Those statistical tables found at the back of statistics textbooks belong in the museum. Statistics you hear from the news and research are produced by computers. This book uses R. You may ask why not using a spreadsheet, such as Microsoft Excel or Google Sheets. While they are excellent tools - easy to use, they have significant shortcomings. In particular, they lack of reproducibility, as they operate through point-clicking, drag-and-drop, etc.

This book intends to provide readers with the bare minimum of R to start analyzing data. In particular, this book is written for readers who have scant experience in computer programming or scripting. I want them to focus on deploying R's rich library of statistical tools to get the desired results rather than being bogged down by the technical details and cryptic syntax of the R language. Although most people think a book like this is about R programming, I prefer to call it **scripting**. The nuance is that the goal of programming is to implement algorithms - a step-by-step procedure to tackle a problem like a cooking recipe. Here we focus on what (expected results) instead of how (algorithms); just tells the R functions here is the data and the parameters, please process them and return the results.

## Usage

The book consists of two parts. Part I focuses on the basics. Chapter 1 discusses R objects and operations act on these objects, descriptive statistical functions, and data wrangling using base R and `tidyverse`. Chapter 2 covers on data visualization. Both base R and `ggplot2` graphical systems will be discussed. Importantly, it provides guidelines to readers how to choose the most suitable graph according to the properties of data. A picture is worth a thousand words. This book will discuss the purpose, pros and cons of each type of graph. Part II includes use cases that leverage the topics discussed in the previous parts.

Scripting is best learned through hands-on practice; it is like learning to swim. You don't just listen to lectures. You've got to jump into the water and

get wet! To try out the examples and exercises, you need to have access to RStudio. There are two options: sign in to the RStudio cloud service from here: (https://posit.cloud/plans?utm_source=Website&utm_medium=IDE_Download), or install R and RStudio on your own computers. Let me use a car analogy to illustrate the relationship between R and RStudio. R is the engine and RStudio is the dashboard. RStudio provides an Interactive Development Environment (IDE) that provides a user-friendly environment for you to unleash the power of R. Both of them work on household Windows and macOS computers. The installation is straightforward; it involves few double-clicks. Here are the links to download R (https://cran.r-project.org/) and RStudio (https://posit.co/download/rstudio-desktop/).

Lastly, this is a live book that I will continue to update with topics to help readers like you. I invite you to check back periodically to discover new material.

# Part I

Part I focuses on the basic R objects and commands, such as atomic and object variables, descriptive statistical functions, and data wrangling using base R and `tidyverse`.

# Basic R

## Learning Objectives:

This chapter covers common basic R functions used for statistical analysis. After finishing this chapter, you should be able to:

- Define R variables.
- Know the differences between an atomic variable and an object variable.
- Perform arithmetic operators.
- Store results in variables.
- Format and print the contents of variables to the console.
- Calculate non-robust and robust statistics.
- Use `seq()` and `with()` functions.

As R scripting is a hands-on topic, I encourage you to open RStudio alongside this book and try out the code to maximize understanding. Let's get started.

## Rules of Naming Variables

The name of an R variable must follow these two rules:

1. It must begin with a letter (A-Z or a-z), a dot "`.`", or an underscore "`_`", and
2. Subsequent characters can be any combinations of letters, digits (0-9), "`.`", or "`_`".

Note that:

- Variable names are case-sensitive, e.g., `volume` and `Volume` are different in the eyes of R.
- Give meaningful names to variables, e.g., `zone2` instead of `z` or even worse `x`. That will help a bunch in understanding the code, even your own code.
- Watch out that some names have been used by R, e.g., `c`, and `head` are function names. However, R is permissive for people to use them as variable names. So, the rule of thumb is to exercise self-discipline not to choose R function names as variable names.

# R Objects

## Atomic and Object Variables

R classifies variables into two broad categories: **atomic type** , and **object**. In the former, the value cannot be split further into smaller R objects. Let us begin with two atomic types: `**numeric**`, and `**character**`.

### Numeric Variables

The simplest atomic type in R is the `numeric` data type. That includes whole numbers, decimal numbers, and numbers represented in scientific notation ($6.023 \times 10^{23}$). In mathematics, an atomic variable is a **scalar**. Here are some examples:

```
# Decimal
height <- 5.6

# Whole number (Integer)
sample_size <- 15

# Scientific notation
avogadro_number <- 6.023e23
```

Don't be mistaken, the `e` in the scientific notation refers to base `10`, e.g., `6.023e23` is $6.023 \times 10^{23}$.

`<-` denotes the assignment operator. Upon execution, the resultant value of the expression on the right-hand side, if no error, is stored in the named variable on the left-hand side. In RStudio, you can see the variable name and its associated value under the **Environment** tab, which usually occupies the top right quadrant of the RStudio desktop.

Note that R supports two equivalent assignment operators. You can use `<-` and `=` interchangeably. I prefer `<-` to `=` as `=` can easily confuse with the equality checking operator `==`.

### Arithmetic Operators

Arithmetic operators that can apply to numeric variables include `+`, `-`, `*`, `/`, `**`, and `^`. The last two operators - `**` and `^` - are exponential; they perform exactly the same function. Here are some examples:

```
radius <- 2.5
diameter <- 2*pi*radius
area <- pi*radius**2
volume <- (4/3)*pi*radius^3
```

Note that `pi` is $\pi$, which is a built-in variable with the value 3.141593.

E.g., the height of an object is 5.6 feet:

```
height <- 5.6
```

### Character Variables

Besides numeric values, R also support operations of non-numeric values, which is called `**character**`. To define a character value, embrace the string of characters with a pair of quotes, either a pair of single or double quotes. E.g.,

```
course <- 'BIOL265'
```

To figure out whether a variable's type is numeric, character, or something else, use the `class()` function,

```
class(course)
```

```
[1] "character"
```

```
class(height)
```

```
[1] "numeric"
```

Obviously, it does not make sense to do math with character type. E.g.,

```
nucleotides*4
```

```
Error:
! object 'nucleotides' not found
```

### Vectors

One commonly used object in R is **`vectors`**. It comprises different components that can be processed collectively as a whole, or individually. A vector can be viewed pictorially as a wagon train. E.g.,

```
# A simple numeric vector for the days in each month
days_of_month <- c(31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31)
```

`days_of_month` is a `vector` object in R. **`c()`** is the function that **combines or collates** several objects, as its name suggests, into a vector object. Elements are indexed or numbered from 1 to the total number of elements in the vector. You can extract a particular element by specifying the index within a pair of square-brackets. E.g.,

```
# Days in February
print(days_of_month[2])
```

```
[1] 28
```

`[2]` means the 2$^{nd}$ element of a vector. We will discuss more about indexing below.

Here is an example of a character vector:

```
nucleotides <- c('A','C','G','T')
```

Noteworthy that all elements held inside a vector must have the same data type (**homogeneous**). For instance, `days_of_month` contains all numeric values, and `nucleotides` contains all character values. If you try to combine objects of different data types into a vector, R will convert them to character type. In the example below, even the first and third values are entered as numeric, R coerces them into character type, the common denominator type, `"1"` and `"0"`.

```
a_vector <- c(1, "success", 0, "fail")
a_vector
```

```
[1] "1"       "success" "0"       "fail"
```

One of the R features that makes arithmetic operations on vectors really convenient is **vectorization**. For instance, if you want to standardize the values of a sample such that the sample mean and sample variance after standardization become 0 and 1, respectively. The formula for standardization is $z_i = \frac{(x_i - \bar{x})}{s}$, where $\bar{x}$ and $s$ are the sample mean and sample standard deviation, respectively. Here's an example

```
heights <- c(5.5, 6.0, 6.1, 5.7, 5.9, 5.6)
x_bar <- mean(heights)
s <- sd(heights)
std_heights <- (heights - x_bar)/s
std_heights
```

```
[1] -1.2677314  0.8451543  1.2677314 -0.4225771  0.4225771 -0.8451543
```

The convenience is that you don't have to specify how to subtract and divide each height individually. Instead, R recognizes the expression `(heights-x_bar)` is to subtract an atomic `x_bar` (scalar) from a vector `heights`. Under the vectorization mechanism, it will subtract `x_bar` from every height in the vector `heights`, similarly for the division by `s`.

`vector` is only one of the many types of objects supported by R, data frame is another commonly used object types, which will be discussed later.

## Factors/Categorical Variables

In R, a categorical variable in statistics is called a **factor**. A categorical variable is further classified into **nominal** or **ordinal**. A nominal variable has no intrinsic order. The `factor()` function is used to define a categorical variable. E.g., a sample of colors is stored in a vector, and you want to define it as a nominal categorical variable.

```
colors <- c("Red","Blue","Blue","Red","Green","Green","Blue","Green","Red","Blue")
colors <- factor(colors, levels=c('Red','Blue','Green'))
colors
```

```
 [1] Red   Blue  Blue  Red   Green Green Blue  Green Red   Blue
Levels: Red Blue Green
```

Categorical variables with intrinsic order are called **ordinal** variables. E.g., the stages of hypertension.

```
patients <- c("Stage 2","Hyperintensive","Stage 2","Hyperintensive","elevated","Stage 1
patients <- factor(patients,
                   levels=c('normal','elevated','Stage 1','Stage 2','Hyperintensive'),
                   ordered = TRUE)
patients
```

```
[1] Stage 2        Hyperintensive Stage 2        Hyperintensive elevated
[6] Stage 2        elevated       Stage 1
Levels: normal < elevated < Stage 1 < Stage 2 < Hyperintensive
```

As you can see above, the hypertension levels are ordered and connected by a less than sign (`<`). ### Data frames

Besides `vector`, another commonly used object is `data frame`. Almost. all the datasets we will use for analysis are in a data frame format. Simply speaking, a data frame is a **tabular** or spreadsheet-like object, like an Excel or Google Sheets. A data frame consists of rows and columns. Each row represents a sample. Each column represents a variable. Rows are expected to have equal length, i.e., the same number of columns. If not, R will either complain or pad the missing columns with a special value `NA` (not available). Let's take a look at an R's built-in data frame `iris`:

```
# Show the first five rows
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

You can peek into the bottom five rows by the `tail()` function.

```
tail(iris)
```

```
    Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
145          6.7         3.3          5.7         2.5 virginica
146          6.7         3.0          5.2         2.3 virginica
147          6.3         2.5          5.0         1.9 virginica
148          6.5         3.0          5.2         2.0 virginica
149          6.2         3.4          5.4         2.3 virginica
150          5.9         3.0          5.1         1.8 virginica
```

As you can see above, the `iris` data frame consists of five variables (columns). Here are the R functions to show the shape of a data frame:

```
# the dimension (row x column) of a data frame.
dim(iris)
```

```
[1] 150   5
```

```
# Only the number of rows
nrow(iris)
```

```
[1] 150
```

```
# Only the number of columns
ncol(iris)
```

```
[1] 5
```

So, we know that `iris` contains 150 rows (flowers) and five columns.

### Accessing Rows and Columns of a Data Frame

Each element in a data frame can be referenced by row and column indexes like the X-Y Cartesian system. Rows are automatically indexed (numbered), where the 1st row is index 1, the 2nd row is index 2, and the index of the last row is the total number of rows, similarly for columns. But often time, columns are named, such as the columns of `iris`; `Sepal.Width`, `Sepal.Length`, etc., as shown in Figure (@ref(fig:Fig1-1)). The two equivalent indexing schemes can occur in parallel.[1]

The syntax of accessing individual elements is `<data frame name>[row, column]`. E.g., the cell at the $5^{th}$ row and $3^{rd}$ column of the iris is:

```
iris[5,3]
```

```
[1] 1.4
```

[1] Rows can be named as well. This is left as an exercise for the readers.

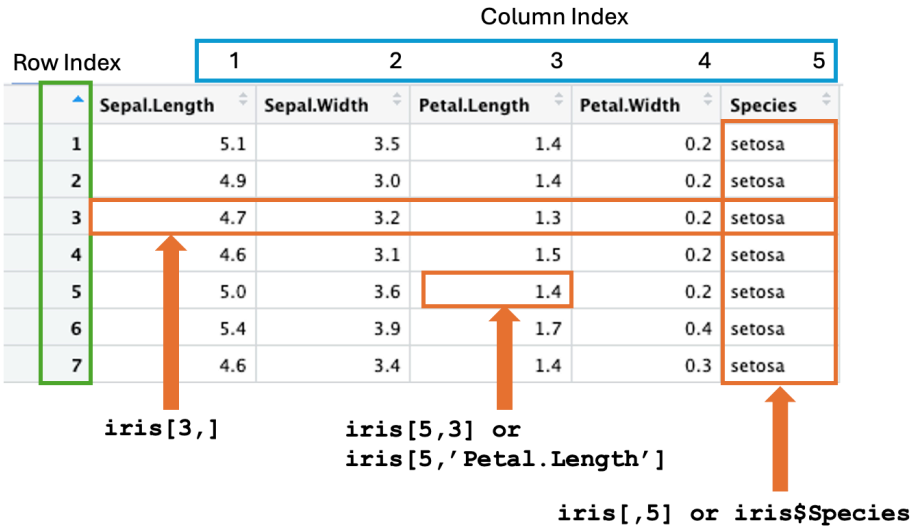



Figure 1.1: Accessing rows and columns of a data frame.

If the column index is ignored, all columns of the specified row(s) are returned as shown in Figure (@ref(fig:Fig1-1)). For example, show all columns of the 3rd row by:

```
# Specify the row number only and leave out the column index.
iris[3,]
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
3          4.7         3.2          1.3         0.2  setosa
```

Similarly, if the row index is skipped, all rows of the specified column(s) are returned (@ref(fig:Fig1-1)). For example,

```
# Specify only the column index.
head(iris[,5])
```

```
[1] setosa setosa setosa setosa setosa setosa
Levels: setosa versicolor virginica
```

R provides an alternative way to access a single column of a data frame. The syntax is `<data frame name>$<column name>`. The equivalence of `iris[,5]` is `iris$Species`:

```
head(iris$Species)
```

```
[1] setosa setosa setosa setosa setosa setosa
Levels: setosa versicolor virginica
```

Note that the `$` method only works for a single column. We will discuss a way shortly how to access multiple columns in one R command.

Besides the numeric column index, columns can be referenced by names as shown in Figure (@ref(fig:Fig1-1)). E.g., the `Petal.Length` of the $5^{th}$ row:

```
iris[5,'Petal.Length']
```

```
[1] 1.4
```

To access more than one column and row, pack the row indices and column names in vectors by the function `c(...)`, which **c**ombines or **c**ollates several items into a vector object. For example, show the `Petal.Length` and `Sepal.Length` of the $123^{th}$ and $96^{th}$ rows:

```
iris[c(123,96),c('Petal.Length','Sepal.Length')]
```

```
    Petal.Length Sepal.Length
123          6.7          7.7
96           4.2          5.7
```

Using row index and column names is the basic accessing method. R provides additional methods to select rows and columns by conditions. More of this topic will be discussed in **Chapter 2: Data Wrangling**.

#### Create a Data Frame

# Descriptive Statistics

## Summary Functions

`length()` shows the number elements in an object. We will use a sample of pumpkin's weights for illustration.

```
# A sample of pumpkins' weights
pumpkins <- c(6.85,6.56,6.23,7.55,4.45,10.14,3.52,6.62,3.76,7.62)
```

The `length()` shows the number of weights defined in `pumpkins`.

```
length(pumpkins)
```

```
[1] 10
```

To be clear, `length()` counts the number of elements in an R object, such as a vector or a data frame. **If the object is atomic, the function will return 1.** For example, in the following example, `dna` is a character (atomic). The function will return 1, not the number of characters.

```
dna <- 'ACGTAGC'
length(dna)
```

```
[1] 1
```

The `range()` prints the minimum and maximum values - numeric and non-numeric - of an object.

```
range(pumpkins)
```

```
[1]  3.52 10.14
```

If you are interested in either the minimum or maximum, here are the corresponding functions:

```
# The lightness
min(pumpkins)
```

```
[1] 3.52
```

```
# The heaviest
max(pumpkins)
```

```
[1] 10.14
```

## Non-robust Statistics

Mean, variance, and standard deviation are non-robust statistics, as they are sensitive to outliers (extreme values).

```
# A sample of pumpkins' weights
pumpkins <- c(6.85,6.56,6.23,7.55,4.45,10.14,3.52,6.62,3.76,7.62)
```

```
# The average weight
mean(pumpkins)
```

```
[1] 6.33
```

```
# The variance
var(pumpkins)
```

```
[1] 4.013489
```

```
# the standard deviation
sd(pumpkins)
```

```
[1] 2.003369
```

## Robust Statistics

Median and quantile are robust statistics, as outliers minimally influence their values.

*Median*

```
# A sample of pumpkins' weights
pumpkins <- c(6.85,6.56,6.23,7.55,4.45,10.14,3.52,6.62,3.76,7.62)
```

```
median(pumpkins)
```

```
[1] 6.59
```

*Quantile*

Standard quantile intervals.

```
quantile(pumpkins)
```

```
    0%    25%    50%    75%   100%
 3.520  4.895  6.590  7.375 10.140
```

A customized quantile intervals.

```
quantile(pumpkins, probs = c(0.1,0.9))
```

```
  10%   90%
3.736 7.872
```

A specific quantile.

```
quantile(pumpkins, probs = c(0.05))
```

```
   5%
3.628
```

```
quantile(pumpkins, probs = c(0.95))
```

```
  95%
9.006
```

**Inter-quantile Region (IQR)**

$$IQR = Q3 - Q1$$

```
IQR(pumpkins)
```

```
[1] 2.48
```

## `summary()` Function

A data frame may consists of many variables of different types in various ranges. Therefore, it makes a lot of sense to take a glimpse of all variables before analysis. It can be done by the `summary()` function to reveal:

1. The variable type of each column.

2. If the column is numeric, a summary of numeric values.
3. If the column is categorical (factor), what are the levels and order, if it is ordinal.

Let's see the summary of `iris`:

```
summary(iris)
```

```
  Sepal.Length    Sepal.Width     Petal.Length    Petal.Width
 Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100
 1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
 Median :5.800   Median :3.000   Median :4.350   Median :1.300
 Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
 3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
 Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
       Species
 setosa    :50
 versicolor:50
 virginica :50
```

The summary will show the range (min. and max.) and quantiles of a numeric column. For a categorical variable (factor), it will show the levels with the number of samples (rows) found for each level, e.g., `Species` is a nominal variable, with 50 samples per species.

# Utilities

## Printing and Formatting

R provides a function called `print()` that prints text to the console. E.g.,

```
print("Hello World!")
```

```
[1] "Hello World!"
```

```
radius <- 2.5
diameter <- 2*pi*radius
print(diameter)
```

```
[1] 15.70796
```

Suppose you want to improve the printout by making it more readable, say `"The diameter of a circle with radius 2.5 cm is 15.70796."` In such a case, you need the format function `paste()`. Here's how to use it:

```
radius <- 2.5
diameter <- 2*pi*radius
print(paste("The diameter of a circle with radius",radius,"cm is",diameter,"."))
```

```
[1] "The diameter of a circle with radius 2.5 cm is 15.707963267949 ."
```

The above printout is almost perfect. If you're picky, you don't like the extra space separating the diameter and the period in the above message, you can use `paste()`'s cousin function `paste0()`. The difference between the two is that `paste0()` will not automatically add an extra space between each component given to the function. Below is the previous print statement with `paste0()`:

```
print(paste0("The diameter of a circle with radius",radius,"cm is",diameter,"."))
```

```
[1] "The diameter of a circle with radius2.5cm is15.707963267949."
```

As you can see, there is no more space between the diameter and the period. But it created other issues; there is no space between "The diameter of a circle with radius" and the actual radius as well. So, when you use `paste0()`, you have to take care of the space separation yourselves. Here is the final version for using `paste0()`:

```
print(paste0("The diameter of a circle with radius ",radius," cm is ",diameter,"."))
```

```
[1] "The diameter of a circle with radius 2.5 cm is 15.707963267949."
```

### `seq()` - create a series of numbers

There are situations where we need a series of equal-interval numbers. For example, override the default breaks - x-ticks - of a histogram, a vector containing all possible outcomes of tossing two dice, i.e, from 2 to 12.

In R, the `seq()` function generates a list of equal-interval numbers, whole and decimal. `seq()` takes the following parameters:

1. `from = 1`, the default is 1.

2. `to = 1`, the default is 1.
3. `by = 1`, the size of the interval between two adjacent numbers. It takes integer and decimal numbers. If it is negative, the series is decreasing.
4. `length.out` specifies how many numbers are returned. This option is an alternative to `by` when you care only about how many numbers are generated instead of the interval between them. See the example below.
5. `along.with`, it generates the same number of numbers according to what is provided to the `along.with` parameter.

**Examples**

By default, `seq()` begins with 1 and has an interval of 1. E.g., generate 10 integers starting with 1.

```
seq(10)
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

Staring from and ending with.

```
seq(-10,10)
```

```
 [1] -10  -9  -8  -7  -6  -5  -4  -3  -2  -1   0   1   2   3   4   5   6   7   8
[20]   9  10
```

Specify the interval.

```
seq(-10,10, by=2)
```

```
 [1] -10  -8  -6  -4  -2   0   2   4   6   8  10
```

A decreasing sequence.

```
seq(10,by=-1)
```

```
 [1] 10  9  8  7  6  5  4  3  2  1
```

Just specify the desired numbers to be generated. As you can see, the interval is pretty precise.

```
seq(-5,7,length.out=20)
```

```
 [1] -5.00000000 -4.36842105 -3.73684211 -3.10526316 -2.47368421 -1.84210526
 [7] -1.21052632 -0.57894737  0.05263158  0.68421053  1.31578947  1.94736842
[13]  2.57894737  3.21052632  3.84210526  4.47368421  5.10526316  5.73684211
[19]  6.36842105  7.00000000
```

Generate a sequence that matches the same number in another sequence.

```
x <- seq(1,5)
cat("x =",x,"\n")
```

```
x = 1 2 3 4 5
```

```
y <- seq(-5, along.with=x)
cat("y =",y,"\n")
```

```
y = -5 -4 -3 -2 -1
```

**Note:** What is the difference between `cat(...)` and `print(paste(...))` you saw above? The main difference is that `print(paste(...))` creates an R object (a text) and then prints it, while `cat(...)` directly writes text to the console without creating a formal R object. See example below you will understand the nuance.

```
msg1 <- print(paste("Hello","World"))
```

```
[1] "Hello World"
```

```
print(msg1)
```

```
[1] "Hello World"
```

```
msg2 <- cat("Hello","World","\n")
```

```
Hello World
```

```
print(msg2)
```

```
NULL
```

Is it really important? No, unless you're a R developer.

### `with()`

It set up the context of a data frame so it spares you from using the `$` to access the columns of a data frame. It is commonly used together with a function applied to one or more columns. E.g., the median of `Petal.Width`. It can done in two ways:

```
# Use $
median(iris$Petal.Width)
```

```
[1] 1.3
```

```
# Or, use with()
with(iris, median(Petal.Width))
```

```
[1] 1.3
```

For the straightforward example above, you do not see the value of `with()`. Let's consider a more complicated row selection example. Find `Sepal.Length` in `iris` where its `Petal.Length` is less than 4 and `Petal.Width` is greater than 1. The `$`-way is:

```
iris$Sepal.Length[iris$Petal.Length<4 & iris$Petal.Width>1]
```

```
[1] 5.2 5.6 5.6 5.5 5.8 5.1
```

Here's the `with()` version that saves typing `iris` repeatedly.

```
with(iris, Sepal.Length[Petal.Length<4 & Petal.Width>1])
```

```
[1] 5.2 5.6 5.6 5.5 5.8 5.1
```

In short, `with()` is English, so it improves the readability of R statements. Second, it saves some typing of the data frame's name. I recommend using `with()` whenever possible.

## Summary

That's a bare minimum of R scripting required for data analysis. In the next chapter, we will focus on operations applied to data frames. Before you leave, it is useful to recap the R functions discussed in this chapter:

Table 1.1: A Summary of R Operators and Functions

| | | | |
|---|---|---|---|
| `<-` | `head()` | `sd()` | `with()` |
| `c()` | `tail()` | `range()` | + |
| `print()` | `dim()` | `min()` | - |
| `paste()` | `nrow()` | `max()` | * |
| `paste0()` | `ncol()` | `median()` | / |
| `cat()` | `summary()` | `quantile()` | ** |
| `class()` | `mean()` | `IQR()` | ^ |
| `factor()` | `var()` | `seq()` | == |

# Chapter 2

# Data Wrangling

Data wrangling refers to the way to view, change, and reorganize data in a data frame. Before we dive into the operations, it is important to know that there are two distinct schools of data wrangling: base R and `tidyverse`, where the latter has contributed substantially in enhancing the process. We will start with the base R version, followed by the `tidyverse`.

## Learning Objectives:

This chapter covers common data wrangling techniques. After finishing this chapter, you should be able to:

- Select rows that fulfill specified conditions.
- Randomly sample rows from a data frame.
- Sort rows in a data frame.
- Select a subset of columns.
- Add and remove columns to an existing data frame.
- Tabulate data.

## 2.1 Base R

### 2.1.1 Select Rows by Condition(s)

The `subset()` function selects rows if they satisfy the condition(s). Here are the three relational operators to support multi-parts conditions:

1. "or" is denoted by `|`
2. "and" is denoted by `&`
3. "not" is denoted by `!`

Note that the operator for checking equality is "=="; two equal signs touching each other.

```
# Choose all setosa iris.
setosa_iris <- subset(iris, Species == 'setosa')

# How many are selected?
print(paste("There are", nrow(setosa_iris), "rows."))
```

```
[1] "There are 50 rows."
```

```
# Peek into the first few rows of the result.
head(setosa_iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

```
# More than one condition. The relational operator is the logical OR, i.e., a vertical
iris_or_eg <- subset(iris, Sepal.Length > 5 | Sepal.Width < 3 )

# How many?
print(paste("There are", nrow(iris_or_eg), "rows."))
```

```
[1] "There are 124 rows."
```

```
head(iris_or_eg)
```

```
   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1           5.1         3.5          1.4         0.2  setosa
6           5.4         3.9          1.7         0.4  setosa
9           4.4         2.9          1.4         0.2  setosa
11          5.4         3.7          1.5         0.2  setosa
15          5.8         4.0          1.2         0.2  setosa
16          5.7         4.4          1.5         0.4  setosa
```

```
# More than one condition. The relational AND operator.
iris_and_eg <- subset(iris, Petal.Length > 6 & Petal.Width < 2 )

# How many?
print(paste("There are", nrow(iris_and_eg), "rows."))
```

```
[1] "There are 2 rows."
```

```
iris_and_eg
```

```
    Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
108          7.3         2.9          6.3         1.8 virginica
131          7.4         2.8          6.1         1.9 virginica
```

```
# No setosa iris
iris_no_setosa_eg <- subset(iris, Species != 'setosa')

# How many?
print(paste("There are", nrow(iris_no_setosa_eg), "rows."))
```

```
[1] "There are 100 rows."
```

```
head(iris_no_setosa_eg)
```

```
   Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
51          7.0         3.2          4.7         1.4 versicolor
52          6.4         3.2          4.5         1.5 versicolor
53          6.9         3.1          4.9         1.5 versicolor
54          5.5         2.3          4.0         1.3 versicolor
55          6.5         2.8          4.6         1.5 versicolor
56          5.7         2.8          4.5         1.3 versicolor
```

### 2.1.2 Select Rows Randomly

It is quite common to randomly sample a subset of rows from a data frame. It takes to two steps to achieve that:

**Step 1.** Create a list (vector) of random row indices by the `sample()` function. Specify the following parameters:

a.) The range of values. For this example, the range is from 1 to the last row of `iris`, which is the return of `nrow(iris)`.

b) How many samples, e.g., `size = 5`.

c) Can a sample be drawn repeatedly? If not, `replace = F`.

```
# seq(nrow(iris)) generates 1, 2, 3, ..., 150.
rnd_idx <- sample(seq(nrow(iris)), size = 5, replace = F)
rnd_idx
```

```
[1]  12  79  24 124   7
```

Step 2: select rows from iris according to the indices in `rnd_idx`.

```
a_random_sample <- iris[rnd_idx,]

a_random_sample
```

```
    Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
12           4.8         3.4          1.6         0.2     setosa
79           6.0         2.9          4.5         1.5 versicolor
24           5.1         3.3          1.7         0.5     setosa
124          6.3         2.7          4.9         1.8  virginica
7            4.6         3.4          1.4         0.3     setosa
```

### 2.1.3 Sort Rows

It is a two-step process similar to the random sampling of rows from the previus subsection. Suppose you want to sort `iris` in `Sepal.Length` in ascending order (small to large).

Step 1: Obtain a list of row indices that indicates the sorting order to the data.

```
sorted_idx <- order(iris$Sepal.Length)
sorted_idx
```

```
  [1]  14   9  39  43  42   4   7  23  48   3  30  12  13  25  31  46   2  10
 [19]  35  38  58 107   5   8  26  27  36  41  44  50  61  94   1  18  20  22
 [37]  24  40  45  47  99  28  29  33  60  49   6  11  17  21  32  85  34  37
 [55]  54  81  82  90  91  65  67  70  89  95 122  16  19  56  80  96  97 100
 [73] 114  15  68  83  93 102 115 143  62  71 150  63  79  84  86 120 139  64
 [91]  72  74  92 128 135  69  98 127 149  57  73  88 101 104 124 134 137 147
[109]  52  75 112 116 129 133 138  55 105 111 117 148  59  76  66  78  87 109
[127] 125 141 145 146  77 113 144  53 121 140 142  51 103 110 126 130 108 131
[145] 106 118 119 123 136 132
```

Step 2: Rearrange the rows in the data frame according to the sorted row indices.

```
# The 5 shortest Sepal Lengths
head(iris[sorted_idx,])
```

```
   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
14          4.3         3.0          1.1         0.1  setosa
9           4.4         2.9          1.4         0.2  setosa
39          4.4         3.0          1.3         0.2  setosa
43          4.4         3.2          1.3         0.2  setosa
42          4.5         2.3          1.3         0.3  setosa
4           4.6         3.1          1.5         0.2  setosa
```

```
# The 5 longest Sepal Lengths
tail(iris[sorted_idx,])
```

```
    Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
106          7.6         3.0          6.6         2.1 virginica
118          7.7         3.8          6.7         2.2 virginica
119          7.7         2.6          6.9         2.3 virginica
123          7.7         2.8          6.7         2.0 virginica
136          7.7         3.0          6.1         2.3 virginica
132          7.9         3.8          6.4         2.0 virginica
```

Note that the above two steps do NOT change the original order of the rows in the data frame. If you intend to change the row order permanently, use the assignment operator to update the existing data frame or create a new data frame to store the sorted rows. See below:

```
ascending_iris <- iris[sorted_idx,]

# The shortest Sepal Length
head(ascending_iris)
```

```
   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
14          4.3         3.0          1.1         0.1  setosa
9           4.4         2.9          1.4         0.2  setosa
39          4.4         3.0          1.3         0.2  setosa
43          4.4         3.2          1.3         0.2  setosa
42          4.5         2.3          1.3         0.3  setosa
4           4.6         3.1          1.5         0.2  setosa
```

```
# The longest Sepal Length
tail(ascending_iris)
```

```
    Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
106          7.6         3.0          6.6         2.1 virginica
118          7.7         3.8          6.7         2.2 virginica
119          7.7         2.6          6.9         2.3 virginica
123          7.7         2.8          6.7         2.0 virginica
136          7.7         3.0          6.1         2.3 virginica
132          7.9         3.8          6.4         2.0 virginica
```

To sort rows in descending order (large to small), add the parameter `decreasing = T` to the `order()` function.

```
desc_idx <- order(iris$Sepal.Length, decreasing = T)
desc_idx
```

```
  [1] 132 118 119 123 136 106 131 108 110 126 130 103  51  53 121 140 142  77
 [19] 113 144  66  78  87 109 125 141 145 146  59  76  55 105 111 117 148  52
 [37]  75 112 116 129 133 138  57  73  88 101 104 124 134 137 147  69  98 127
 [55] 149  64  72  74  92 128 135  63  79  84  86 120 139  62  71 150  15  68
 [73]  83  93 102 115 143  16  19  56  80  96  97 100 114  65  67  70  89  95
 [91] 122  34  37  54  81  82  90  91   6  11  17  21  32  85  49  28  29  33
[109]  60   1  18  20  22  24  40  45  47  99   5   8  26  27  36  41  44  50
[127]  61  94   2  10  35  38  58 107  12  13  25  31  46   3  30   4   7  23
[145]  48  42   9  39  43  14
```

```
decreasing_iris <- iris[desc_idx,]
head(decreasing_iris)
```

```
    Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
132          7.9         3.8          6.4         2.0 virginica
118          7.7         3.8          6.7         2.2 virginica
119          7.7         2.6          6.9         2.3 virginica
123          7.7         2.8          6.7         2.0 virginica
136          7.7         3.0          6.1         2.3 virginica
106          7.6         3.0          6.6         2.1 virginica
```

```
tail(decreasing_iris)
```

```
   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
48          4.6         3.2          1.4         0.2  setosa
```

```
42          4.5         2.3          1.3         0.3  setosa
9           4.4         2.9          1.4         0.2  setosa
39          4.4         3.0          1.3         0.2  setosa
43          4.4         3.2          1.3         0.2  setosa
14          4.3         3.0          1.1         0.1  setosa
```

### 2.1.4 Rearranging Columns

Suppose you want to move the `Species` column to the leftmost position. It can be done using two approaches.

```
# Reorder columns by name
new_iris <- iris[,c("Species","Sepal.Length","Sepal.Width","Petal.Length","Petal.Width")]
head(new_iris)
```

```
  Species Sepal.Length Sepal.Width Petal.Length Petal.Width
1  setosa          5.1         3.5          1.4         0.2
2  setosa          4.9         3.0          1.4         0.2
3  setosa          4.7         3.2          1.3         0.2
4  setosa          4.6         3.1          1.5         0.2
5  setosa          5.0         3.6          1.4         0.2
6  setosa          5.4         3.9          1.7         0.4
```

```
# Reorder columns by index
new_iris <- iris[,c(5,1,2,3,4)]
head(new_iris)
```

```
  Species Sepal.Length Sepal.Width Petal.Length Petal.Width
1  setosa          5.1         3.5          1.4         0.2
2  setosa          4.9         3.0          1.4         0.2
3  setosa          4.7         3.2          1.3         0.2
4  setosa          4.6         3.1          1.5         0.2
5  setosa          5.0         3.6          1.4         0.2
6  setosa          5.4         3.9          1.7         0.4
```

Note that if a data frame contains many columns, say >100, it is daunting to use this approach. A much better way can be done using `dplyr`, so stay tuned.

### 2.1.5 Focus on a Subset of Columns

If a data frame contains many columns, way more than you need for analysis. It is neat to focus on those columns and remove the rest. Let's suppose you want to focus only on sepal information and species.

```
sepal_iris <- iris[,c("Sepal.Length","Sepal.Width","Species")]
dim(sepal_iris)
```

```
[1] 150   3
```

```
head(sepal_iris)
```

```
  Sepal.Length Sepal.Width Species
1          5.1         3.5  setosa
2          4.9         3.0  setosa
3          4.7         3.2  setosa
4          4.6         3.1  setosa
5          5.0         3.6  setosa
6          5.4         3.9  setosa
```

### 2.1.6 Add a New Column

Suppose we want to add a new column called `Sepal.Ratio`, which is the ratio `Sepal.Length` to `Sepal.Width`. Here is the code:

```
iris$Sepal.Ratio <- iris$Sepal.Length/iris$Sepal.Width
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species Sepal.Ratio
1          5.1         3.5          1.4         0.2  setosa    1.457143
2          4.9         3.0          1.4         0.2  setosa    1.633333
3          4.7         3.2          1.3         0.2  setosa    1.468750
4          4.6         3.1          1.5         0.2  setosa    1.483871
5          5.0         3.6          1.4         0.2  setosa    1.388889
6          5.4         3.9          1.7         0.4  setosa    1.384615
```

You can see the new column `Sepal.Ratio` is added to the rightmost position.

### 2.1.7 Remove a Column

You can remove a column from a data frame, e.g., remove the `Sepal.Ratio` added above.

```
iris$Sepal.Ratio <- NULL

# the column is gone
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

### 2.1.8 Tabulation

Tally the number of samples per one or more categorical variables (factors) by the `table()` function.

```
table(iris$Species)
```

```
    setosa versicolor  virginica
        50         50         50
```

```
# Alternatively, use with()
with(iris, table(Species))
```

```
Species
    setosa versicolor  virginica
        50         50         50
```

It can tally more than one categorical variable. Let's us another built-in dataset `CO2` to illustrate the point.

```
with(CO2, table(Plant, Type))
```

```
     Type
Plant Quebec Mississippi
  Qn1      7           0
  Qn2      7           0
  Qn3      7           0
  Qc1      7           0
  Qc3      7           0
  Qc2      7           0
  Mn3      0           7
  Mn2      0           7
  Mn1      0           7
  Mc2      0           7
  Mc3      0           7
  Mc1      0           7
```

## 2.2 tidyverse

There is an alternate, and importantly, less cryptic way to perform data wrangling. This new set of functions are offered by the R package `dplyr`. One design goal of `dplyr` is to use **verbs** to name these operations, a huge difference from the unpronounceable symbols, such as `$`, `[`, etc. by base R as shown in the previous section.

Before we dive into `dplyr`, it is noteworthy to mention that there is a slew of packages with the aim of improving data cleaning, data visualization, and loading data in various formats. To spare people from the question of when to use what package, all these packages are bundled in a single package called `tidyverse`. Once you have activated `tidyverse`, `dplyr`, `ggplot` (for plotting data), and other packages are available for use. When you call a function, you don't have to worry if it is from `dplyr` or `tidyr` because the `tidyverse` community has resolved all name conflicts. An analogy is like there is only one "Main Street" in `tidyverse` package; no ambiguity at all. **The take home is: just activate `tidyverse` and you are good to go.**

`tidyverse` does not come with the base R so you have to install it manually. But the installation is simple, click the menu option Tools, select the first option "install packages", type "tidyverse", and click the "Install" button.

Let's activate `tidyverse` by using the `library()` function. Alternatively, you can use `require()`; they are largely doing the same thing. Note that you only need to activate a package once per session.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to
```

The `library()` function has churned out many messages telling you what packages (9 of them) are loaded together.

Let's use `tidyverse` to repeat the data wrangling tasks by base R above.

### 2.2.1 Select Rows by Condition(s)

The function is `filter()`. It chooses rows that satisfy the specified condition(s). The syntax is `filter(data_frame, conditions)`. E.g.,

```
# select all setosa iris
setosa_iris <- filter(iris, Species == 'setosa')

# How many?
print(paste("There are", nrow(setosa_iris), "rows."))
```

```
[1] "There are 50 rows."
```

```
# More than one condition. The symbol for the logical OR operator is a vertical bar |
iris_or_eg <- filter(iris, Sepal.Length > 5 | Sepal.Width < 3 )

# How many?
print(paste("There are", nrow(iris_or_eg), "rows."))
```

```
[1] "There are 124 rows."
```

```
head(iris_or_eg)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          5.4         3.9          1.7         0.4  setosa
3          4.4         2.9          1.4         0.2  setosa
4          5.4         3.7          1.5         0.2  setosa
5          5.8         4.0          1.2         0.2  setosa
6          5.7         4.4          1.5         0.4  setosa
```

```
# More than one condition. The symbol for the logical AND operator is &
iris_and_eg <- filter(iris, Petal.Length > 6 & Petal.Width < 2 )

# How many?
print(paste("There are", nrow(iris_and_eg), "rows."))
```

```
[1] "There are 2 rows."
```

```
iris_and_eg
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
1          7.3         2.9          6.3         1.8 virginica
2          7.4         2.8          6.1         1.9 virginica
```

```
# No setosa iris
iris_not_eg <- filter(iris, Species != 'setosa')

# How many?
print(paste("There are", nrow(iris_not_eg), "rows."))
```

```
[1] "There are 100 rows."
```

```
head(iris_not_eg)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
1          7.0         3.2          4.7         1.4 versicolor
2          6.4         3.2          4.5         1.5 versicolor
3          6.9         3.1          4.9         1.5 versicolor
4          5.5         2.3          4.0         1.3 versicolor
5          6.5         2.8          4.6         1.5 versicolor
6          5.7         2.8          4.5         1.3 versicolor
```

### 2.2.2 Select Rows Randomly

With `tidyverse`, you can do it in one step by using the `sample_n()` function.

```
a_random_sample <- sample_n(iris, size = 5)

a_random_sample
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
1          5.8         2.7          5.1         1.9  virginica
2          4.9         2.4          3.3         1.0 versicolor
3          4.6         3.1          1.5         0.2     setosa
4          5.8         2.7          5.1         1.9  virginica
5          5.5         4.2          1.4         0.2     setosa
```

### 2.2.3 Sort Rows in a Data Frame

Use `arrange()`. Note that the function will not change the original data frame. Use the assignment operator (`<-`) to change the original data frame or store the changes in a new data frame. For example, sort `iris` by `Sepal.Length`, and store the sorted result in a new data frame.

```
sorted_iris <- arrange(iris, Sepal.Length)

# The shortest Sepal Length
head(sorted_iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          4.3         3.0          1.1         0.1  setosa
2          4.4         2.9          1.4         0.2  setosa
3          4.4         3.0          1.3         0.2  setosa
4          4.4         3.2          1.3         0.2  setosa
5          4.5         2.3          1.3         0.3  setosa
6          4.6         3.1          1.5         0.2  setosa
```

```
# The longest Sepal Length
tail(sorted_iris)
```

```
    Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
145          7.6         3.0          6.6         2.1 virginica
146          7.7         3.8          6.7         2.2 virginica
147          7.7         2.6          6.9         2.3 virginica
148          7.7         2.8          6.7         2.0 virginica
149          7.7         3.0          6.1         2.3 virginica
150          7.9         3.8          6.4         2.0 virginica
```

To sort rows according to the descending order of the sorting column, simply mark the sorting column with a minus -. E.g.,

```
decreasing_iris <- arrange(iris, -Sepal.Length)
head(decreasing_iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
1          7.9         3.8          6.4         2.0 virginica
2          7.7         3.8          6.7         2.2 virginica
3          7.7         2.6          6.9         2.3 virginica
4          7.7         2.8          6.7         2.0 virginica
5          7.7         3.0          6.1         2.3 virginica
6          7.6         3.0          6.6         2.1 virginica
```

```
tail(decreasing_iris)
```

```
    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
145          4.6         3.2          1.4         0.2  setosa
```

```
146          4.5         2.3          1.3         0.3  setosa
147          4.4         2.9          1.4         0.2  setosa
148          4.4         3.0          1.3         0.2  setosa
149          4.4         3.2          1.3         0.2  setosa
150          4.3         3.0          1.1         0.1  setosa
```

Specify more than one sorting column is much easier in `dplyr` than base R. E.g. sort `iris` in descending order `Sepal.Length` and ascending order of `Petal.Length`.

```
double_sorted_iris <- arrange(iris, -Sepal.Length, Petal.Length)
head(double_sorted_iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
1          7.9         3.8          6.4         2.0 virginica
2          7.7         3.0          6.1         2.3 virginica
3          7.7         3.8          6.7         2.2 virginica
4          7.7         2.8          6.7         2.0 virginica
5          7.7         2.6          6.9         2.3 virginica
6          7.6         3.0          6.6         2.1 virginica
```

```
tail(double_sorted_iris)
```

```
    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
145          4.6         3.1          1.5         0.2  setosa
146          4.5         2.3          1.3         0.3  setosa
147          4.4         3.0          1.3         0.2  setosa
148          4.4         3.2          1.3         0.2  setosa
149          4.4         2.9          1.4         0.2  setosa
150          4.3         3.0          1.1         0.1  setosa
```

For rows with equal `Sepal.Length` (rows 2-5, 147-149), you can see their `Petal.Length` are in ascending order.

### 2.2.4 Rearranging columns of a data frame

Suppose you want to move the `Species` column to the leftmost position. You're going to love `dplyr`, as it can be done easily by using two functions: `select()` and `everything()`.

```
# Reorder columns by name
new_iris <- select(iris, Species, everything())
head(new_iris)
```

```
  Species Sepal.Length Sepal.Width Petal.Length Petal.Width
1  setosa          5.1         3.5          1.4         0.2
2  setosa          4.9         3.0          1.4         0.2
3  setosa          4.7         3.2          1.3         0.2
4  setosa          4.6         3.1          1.5         0.2
5  setosa          5.0         3.6          1.4         0.2
6  setosa          5.4         3.9          1.7         0.4
```

### 2.2.5 Focus on a Subset of Columns

If a data frame contains many columns, way more than you need for analysis. It is neat to focus on those columns and remove the rest. E.g., use the `select()` function to keep only sepal information and species.

```
sepal_iris <- select(iris, Sepal.Length, Sepal.Width, Species)
dim(sepal_iris)
```

```
[1] 150   3
```

```
head(sepal_iris)
```

```
  Sepal.Length Sepal.Width Species
1          5.1         3.5  setosa
2          4.9         3.0  setosa
3          4.7         3.2  setosa
4          4.6         3.1  setosa
5          5.0         3.6  setosa
6          5.4         3.9  setosa
```

### 2.2.6 Add a New Column

Use the `mutate()` function add columns to a data frame.

```
iris <- mutate(iris, Sepal.Ratio = Sepal.Length/Sepal.Width)
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species Sepal.Ratio
1          5.1         3.5          1.4         0.2  setosa    1.457143
2          4.9         3.0          1.4         0.2  setosa    1.633333
3          4.7         3.2          1.3         0.2  setosa    1.468750
4          4.6         3.1          1.5         0.2  setosa    1.483871
5          5.0         3.6          1.4         0.2  setosa    1.388889
6          5.4         3.9          1.7         0.4  setosa    1.384615
```

### 2.2.7 Remove a Column

Use the `select()` function as before but put a minus `'-'` symbol before the column name to be removed.

```
iris <- select(iris, -Sepal.Ratio)
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

### 2.2.8 Rename a Column

So far, you have seen all the data wrangling tasks performed by base R using `tidyverse`. Starting from here, you are going to demonstrate tasks that are harder to do with base R. The first task is to rename a column. Suppose you want to rename `Species` to `Iris_Species`. You can do it by the `rename()` function:

```
# new_name = existing_name
iris <- rename(iris, Iris_Species = Species)
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Iris_Species
1          5.1         3.5          1.4         0.2       setosa
2          4.9         3.0          1.4         0.2       setosa
3          4.7         3.2          1.3         0.2       setosa
4          4.6         3.1          1.5         0.2       setosa
5          5.0         3.6          1.4         0.2       setosa
6          5.4         3.9          1.7         0.4       setosa
```

Let's revert the renaming so it will not affect the following examples.

```
iris <- rename(iris, Species = Iris_Species)
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
```

```
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

### 2.2.9 Pipe

Often, data wrangling is a multi-step process in which many intermediate steps are involved before producing the final result. E.g., select iris where its `Petal.Length` is between 0.5 and 2, and sort the results in descending order of `Petal.Length`. As you can see, it involves filtering, followed by sorting. It can done into two steps via an intermediate data frame as below:

```
tmp_iris <- filter(iris, Petal.Length >= 0.5 & Petal.Length <=2)
final_iris <- arrange(tmp_iris, -Petal.Length)
dim(final_iris)
```

```
[1] 50  5
```

```
head(final_iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          4.8         3.4          1.9         0.2  setosa
2          5.1         3.8          1.9         0.4  setosa
3          5.4         3.9          1.7         0.4  setosa
4          5.7         3.8          1.7         0.3  setosa
5          5.4         3.4          1.7         0.2  setosa
6          5.1         3.3          1.7         0.5  setosa
```

The final result contains 50 rows.

`tidyverse` provides a neater way - no intermediate data frames - to do it by the concept of **pip**. You can imagine it like the assembly line of a factory, where an upstream work station passes (pipe) intermediate results to the next work station for further processing.

The pip symbol is represented by **%>%**, no space between them. Let's see how to apply pip to the above example without creating the intermediate data frame `tmp_iris`.

```
final_results <- iris %>%
                   filter(Petal.Length >= 0.5 & Petal.Length <=2) %>%
                   arrange(-Petal.Length)
dim(final_iris)
```

```
[1] 50  5
```

```
head(final_iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          4.8         3.4          1.9         0.2  setosa
2          5.1         3.8          1.9         0.4  setosa
3          5.4         3.9          1.7         0.4  setosa
4          5.7         3.8          1.7         0.3  setosa
5          5.4         3.4          1.7         0.2  setosa
6          5.1         3.3          1.7         0.5  setosa
```

Let's walk through the code above line by line.

- `iris %>%` sends all rows, 150 in total, to the downstream `filter()` function.
- `filter(Petal.Length >= 0.5 & Petal.Length <=2) %>%` receives the data, and check each row if it satisfies the specified conditions. If so, send it to the downstream process; otherwise do not pass down.
- `arrange(-Petal.Length)` received rows that satisfy the previous filtering process, and sort them in descending order of `Petal.Length`.
- `final_results <-` the sorted result is assigned to the variable `final_results`.

By using pip (`%>%`), you can avoid intermediate variables, making the code cleaner and easily to read.

### 2.2.10 Tabulation and Summarize

Another task which is useful but difficult to get it done in base R is to produce summary information by group. E.g., you want to produce a table that shows the average sepal length for each species. Here's how to do it using two new functions `group_by()` and `summarize()`, and pipe them together.

E.g., tally the number of samples by `Plant` and 'Type' in `CO2`. The function that does the tallying is `n()`.

```
CO2 %>%
  group_by(Plant, Type) %>%
  summarize(n=n())
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by Plant and Type.
i Output is grouped by Plant.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(Plant, Type))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 12 x 3
# Groups:   Plant [12]
   Plant Type            n
   <ord> <fct>       <int>
 1 Qn1   Quebec          7
 2 Qn2   Quebec          7
 3 Qn3   Quebec          7
 4 Qc1   Quebec          7
 5 Qc3   Quebec          7
 6 Qc2   Quebec          7
 7 Mn3   Mississippi     7
 8 Mn2   Mississippi     7
 9 Mn1   Mississippi     7
10 Mc2   Mississippi     7
11 Mc3   Mississippi     7
12 Mc1   Mississippi     7
```

You do not see much difference between base R and `tidyverse` in the example above. But if you want to do more than tallying, `summarize()` is a treat, as `summarize()` supports the descriptive statistics functions mentioned before, such as mean(), sd(), median(), quantile(), etc. E.g,

```
iris %>%
  group_by(Species) %>%
  summarize(u=mean(Sepal.Length))
```

```
# A tibble: 3 x 2
  Species        u
  <fct>      <dbl>
1 setosa      5.01
2 versicolor  5.94
3 virginica   6.59
```

You can apply multiple functions in `summarize()`. E.g., shows the number of samples and standard deviation.

```
iris %>%
  group_by(Species) %>%
  summarize(n=n(),
            u=mean(Sepal.Length),
            s=sd(Sepal.Length))
```

```
# A tibble: 3 x 4
  Species        n     u     s
  <fct>      <int> <dbl> <dbl>
1 setosa        50  5.01 0.352
2 versicolor    50  5.94 0.516
3 virginica     50  6.59 0.636
```

You can group by more than one column. Let's switch to `CO2` dataset to illustrate this point.

```
CO2 %>%
  group_by(Plant, Type, Treatment) %>%
  summarize(n=n(),
            u=mean(conc),
            s=sd(conc))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by Plant, Type, and Treatment.
i Output is grouped by Plant and Type.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(Plant, Type, Treatment))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 12 x 6
# Groups:   Plant, Type [12]
   Plant Type        Treatment      n     u     s
   <ord> <fct>       <fct>      <int> <dbl> <dbl>
 1 Qn1   Quebec      nonchilled     7   435  318.
 2 Qn2   Quebec      nonchilled     7   435  318.
 3 Qn3   Quebec      nonchilled     7   435  318.
 4 Qc1   Quebec      chilled        7   435  318.
 5 Qc3   Quebec      chilled        7   435  318.
 6 Qc2   Quebec      chilled        7   435  318.
 7 Mn3   Mississippi nonchilled     7   435  318.
 8 Mn2   Mississippi nonchilled     7   435  318.
 9 Mn1   Mississippi nonchilled     7   435  318.
10 Mc2   Mississippi chilled        7   435  318.
11 Mc3   Mississippi chilled        7   435  318.
12 Mc1   Mississippi chilled        7   435  318.
```

**Two remarks:**

1. `group_by()` is never used alone. It is often followed by `summarize()`.
2. The variables feed to `group_by()` must be factor, i.e., categorical.

### 2.2.11 A Summary of `tidyverse` functions

- `filter()`: select rows that satistify the specified conditions.
- `select()`: choose or drop certain columns.
- `arrange()`: sort rows by the specified columns.
- `rename()`: change the name of a column.
- `mutate()`: add a column or modify the values of an existing column.
- `group_by()`, followed by `summarize()`
- pip `%>%`: funnel intermediate result from an upstream process to the input of an immediate downstream process.

## 2.3 Summary

In this chapter, you have seen how to perform data wrangling with basic R and `tidyverse`. What you have seen is only a fraction of what `tidyverse` can do. Readers who want to explore the full capability of `tidyverse` should consult the most authorative source: **R for Data Science** by Hadley Wickham & Garrett Grolemund (https://r4ds.had.co.nz/). In the next chapter, you will see how to plot data. A picture is worth a thousand words. Data visualization is an indispensible step for data analysis. But before the next topic, let's recall all the new R commands discussed in this chapter.

Table 2.1: A Summary of R Operators and Functions

| | | | |
|---|---|---|---|
| \| | `sample()` | `filter()` | `summarize()` |
| `&` | `order()` | `select()` | `n()` |
| `==` | `NONE` (a keyword) | `arrange()` | pip `%>%` |
| `!` | `table()` | `rename()` | |
| `!=` | `library()` | `mutate()` | |
| `subset()` | `require()` | `group_by()` | |

# Chapter 3

# Data Visualization

After laboring yourselves to collect, clean, and analyze the data, you want to share the findings and engage the audience with insights about the significance of the study. But most people are not good at extracting insights from rows and columns of numbers. It is the job of the researchers to get important messages across to the audience so they can understand them with as little effort as possible.

## Learning Objectives:

This chapter discusses the properties of various graphs, what they are good at and not so good at, and how to plot them in R using two different graphical systems. After finishing this chapter, you should be able to:

- Choose the most appropriate graph based on the number of variables and their types.
- Convey the message you want to convey to the audience embedded in the data through graphs.
- Decorate graphs with labels, a main title, color, and other graphical attributes using the base R graphical system.
- Create publication-quality graphs using the `ggplot2` package.

## 3.1 Types of Graphs

The best way to choose the most appropriate graph depends on the data. Specially, the types of variables (numeric or categorical) and how many of them.

You will plot graph for:

1. One numeric variable.
2. One categorical variable.
3. One categorical and one numeric variables.
4. Two or more numeric variables.
5. A summary table.

## 3.2 Base R Graphics

This graphical system is built into the base R; no additional package is required. It is simple, ideal for creating graphs promptly for prototyping purposes and get the job done. The flip side is that the graphs may appear a bit primitive. Some features are hard to be implemented. Anyway, if plotting graphs using R is new to you, base R is a great starting point. You will understand the basic characteristics of different graphs without requiring too much technical details.

### 3.2.1 One Numeric Variable

You can plot a histogram of the numeric variable to see its distribution, such as the spread, the peak, symmetry, and flatness. If your concern is robust statistics, a boxplot will do the job.

#### 3.2.1.1 Histogram

Let's examine the distribution of `iris`'s `Sepal.Width`.

```
par(mfrow=c(2,2))

# Top left
hist(iris$Sepal.Width)

# Top right
hist(iris$Sepal.Width,
     xlab="Sepal Width",
     ylab = "Count",
     main = "Histogram of Sepal Width")

# Bottom left
```

```r
hist(iris$Sepal.Width,
     col = 'skyblue',
     border = 'white',
     xlab="Sepal Width",
     ylab = "Count",
     main = "Color options")

# Bottom right
hist(iris$Sepal.Width,
     breaks = seq(2,4.5,by=0.1),
     col = 'skyblue',
     border = 'white',
     xlab="Sepal Width",
     ylab = "Count",
     main = "breaks option")
```

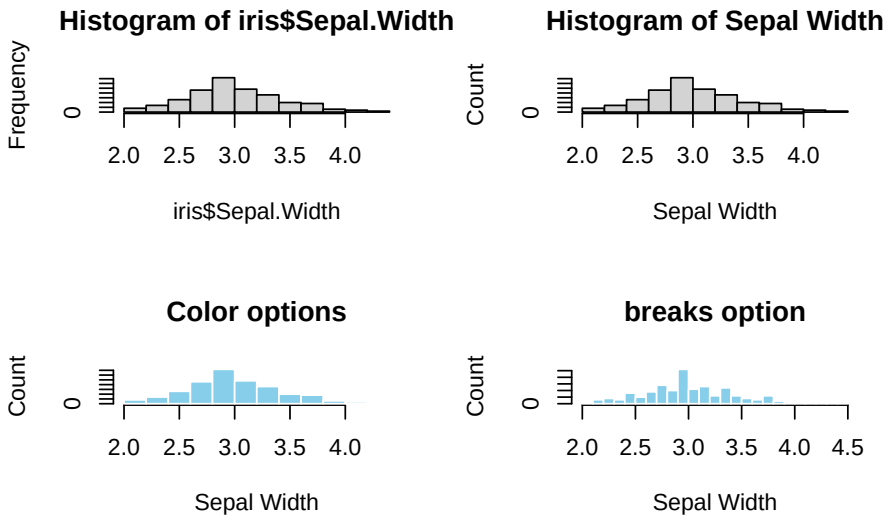



Figure 3.1: Base R Histogram. Top left: default. Top right: Customized X-Y labels and the main title. Bottom left: change color of bars and border. Bottom right: change the bin width (`breaks=`)

For the bottom-right graph, the `seq()` function (@ref(seq-function)) is used to generate a list of numbers for the `breaks` parameter that represents x-ticks.

Note: `par(mfrow=c(2,2))` splits the plotting area into a 2x2 grid such that multiple graphs can be grouped

#### 3.2.1.2 Boxplot

If the message is robust statistics, such as median and quantiles, or the presence of outliers, the best graph is a boxplot. The y-axis represents the numeric variable, and the x-axis has no meaning.

```
par(mfrow=c(2,2))

# Top left
boxplot(iris$Sepal.Width)

# Top right
boxplot(iris$Sepal.Width,
        ylab="Sepal Width",
        main="Boxplot of Sepal Width")

# Bottom left
boxplot(iris$Sepal.Width,
            col = "skyblue",
            ylab="Sepal Width",
            outcol = 'red',
            outpch = 19,
            main="Color options")

# Bottom right
boxplot(iris$Sepal.Width,
        horizontal = TRUE,
        xlab="Sepal Width",
        main="Horizontal Boxplot")
```

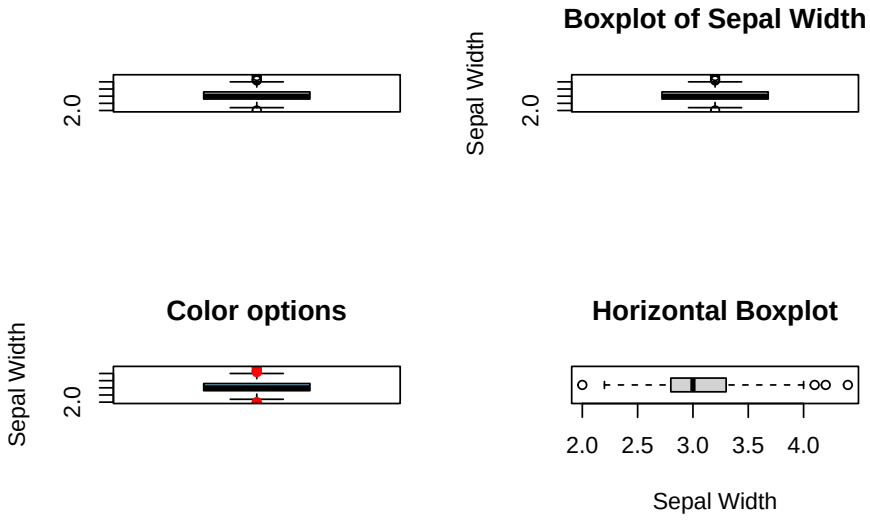



Figure 3.2: Boxplot. Top left: Default, Top right: labels and main title. Bottom left: color options. Bottom right: `horizontal = TRUE`

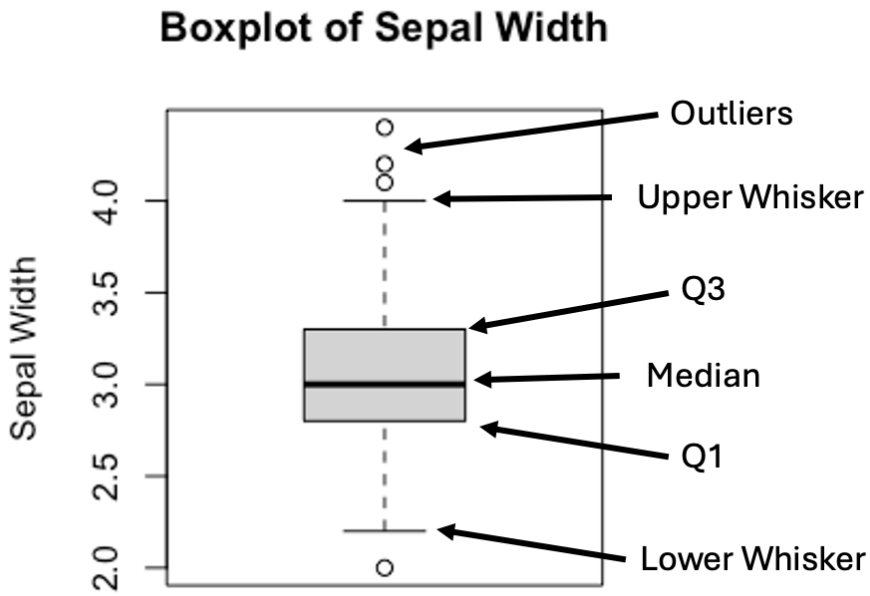



Figure 3.3: Anatomy of a Boxplot

The upper and lower whiskers are defined as:

$$Upper\ Whisker = Q3 + 1.5 \times IQR$$

$$Lower\ Whisker = Q1 - 1.5 \times IQR$$

, where $IQR = (Q3 - Q1)$.

In practice, values that are greater or less than the upper or lower whiskers are treated as outliers.

Histogram offers a glimpse of how the data is distributed, displaying the peak(s), and long tail, if any, that is less obvious in a boxplot. However, pinpointing the median and outliers on a histogram is non-trivial. Therefore, it makes sense to combine the strengths of both plots in a single graph. To be sure the two plots are comparable, make sure to set the axes in the same range by the `xlim` and/or `ylim=` parameters. Here's an example below:

```
par(mfrow=c(2,1))

hist(iris$Sepal.Width,
     breaks = seq(2,4.5,by=0.1),
     col = 'skyblue',
     border = 'white',
     xlim = c(2,4.5),
     xlab = "",
     ylab = "Count",
     main = "Stacking Up Graphs")

boxplot(iris$Sepal.Width,
        horizontal = TRUE,
        ylim = c(2,4.5),
        xlab="Sepal Width",
        main="")
```

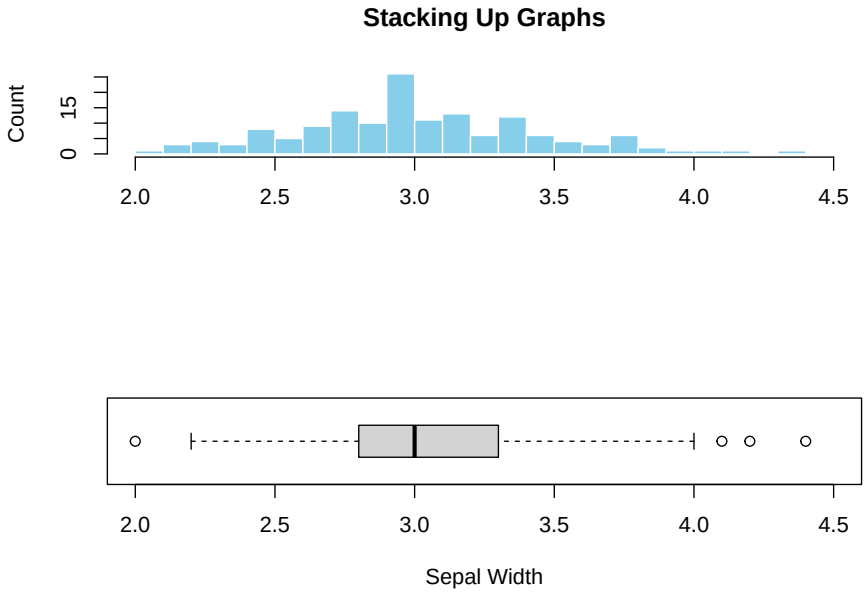



Figure 3.4: Stacking up two graphs.

Figure (@ref(fig:Fig3-4))) illustrated the added advantage of combining graphs. It is clear that the peak in the histogram is almost perfectly aligned with the median shown in the boxplot, indicating symmetric data distribution. Additionally, the outliers hardly visible from the histogram can easily be spotted in the boxplot. This example illustrates the synergy and additional perspectives created by combining different graphs.

### 3.2.2 One Categorical Variable

When it comes to categorical variables, we are usually interested in the number of samples fall in each category of a categorical variable. As discussed in Chapter 2, the tallying can be done using the `table()` function (@ref(base-tabulation)) or `group_by()` followed by `summarize(n=n())` (@ref(tidyverse-tabulation)).

#### 3.2.2.1 Barplot

Barplot is the standard choice for visualizing tallied data. The only input to the R's `barplot()` graph function is a tabulated count of the categorical variable. Here's an example:

```r
par(mfrow=c(2,2))

# Top left
barplot(with(iris, table(Species)))

# Top right
barplot(with(iris, table(Species)),
        xlab="Species",
        ylab="Count",
        main="A Barplot of Species")

# Bottom left
barplot(with(iris, table(Species)),
        col='skyblue',
        border='blue',
        xlab="Species",
        ylab="Count",
        main="Color Options")

# Bottom right
barplot(with(iris, table(Species)),
        horiz = TRUE,
        col='skyblue',
        border='blue',
        xlab="Count",
        ylab="Species",
        main="A Horizontal Barplot")
```

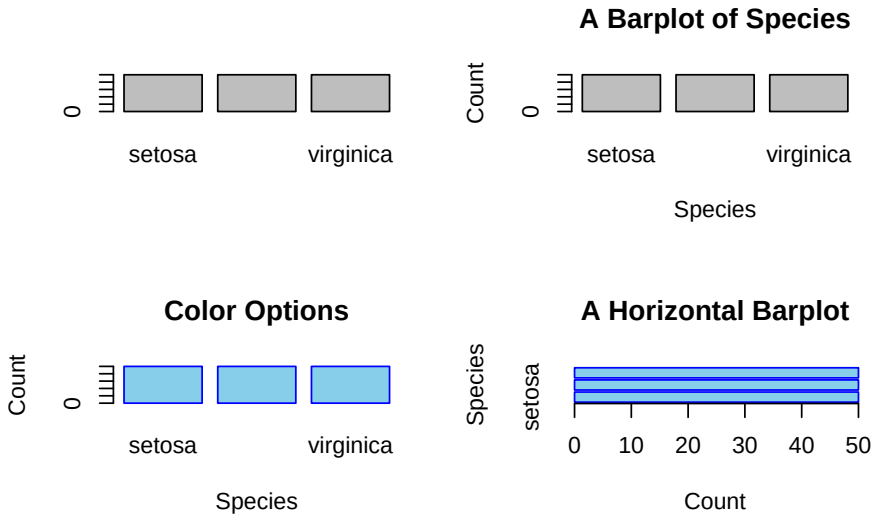



Figure 3.5: A simple bar plot. Top left: Default. Top right: with tables

Sometime you might want to reorder the categories along the x- or y-axis. By default, `barplot()` places the bars from left to right according to the current order of the levels. E.g., the levels of `iris$Species` is:

```
levels(iris$Species)
```

```
[1] "setosa"     "versicolor" "virginica"
```

You can change the current order by updating the factor variable by the `factor()` function (@ref(factor-levels)). Suppose, the desired order is `virginica`, `setosa`, and `veriscolor`.

```
iris$Species <- factor(iris$Species,
                       levels = c("virginica", "setosa", "versicolor"),
                       ordered = TRUE)
levels(iris$Species)
```

```
[1] "virginica"  "setosa"     "versicolor"
```

```
barplot(with(iris, table(Species)),
        col='skyblue',
        border='blue',
```

```
        xlab="Species",
        ylab="Count",
        main="Reordered Species")
```

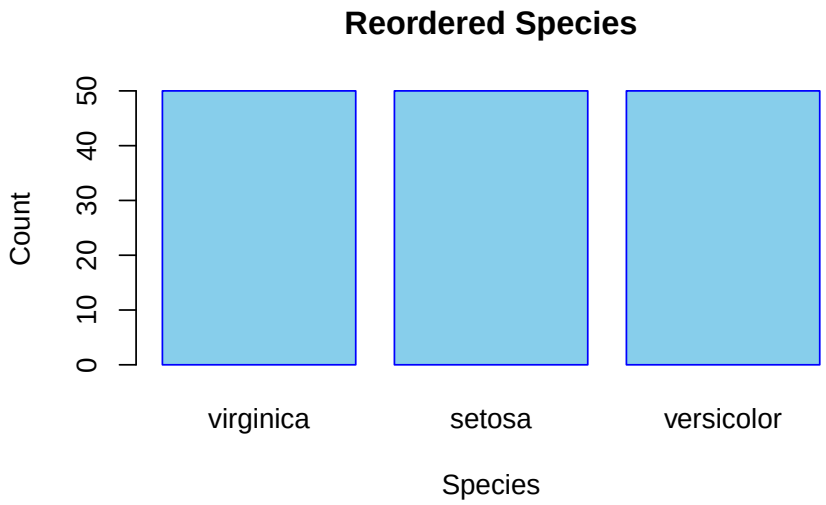



Figure 3.6: Reorder the Bars of a Barplot

#### 3.2.2.2 Pie Chart

Besides a barplot, a pie chart is also a good fit for displaying tallied data.

```
par(mfrow=c(2,2))

# Top left
pie(with(iris, table(Species)))

# Top left
pie(with(iris, table(Species)),
    col = c('skyblue', 'orange', 'white'),
    main="Species")

# Bottom right
counts <- with(iris, table(Species))
my_labs <- paste0(names(counts),"(",counts,")")
pie(counts,
```

```
    labels = my_labs,
    col = c('skyblue', 'orange', 'white'),
    main = "Species Counts")


# Bottom right
counts <- with(iris, table(Species))
fractions <- round(counts/sum(counts),2)
my_labs <- paste0(names(counts),"(",fractions,")")
pie(counts,
    labels = my_labs,
    col = c('skyblue', 'orange', 'white'),
    main = "Species Fractions")
```

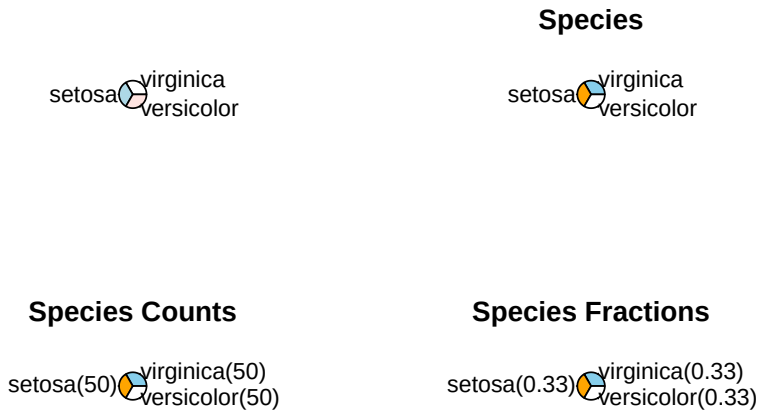



Figure 3.7: A simple pie chart. Top left: Default. Top right: with tables

### 3.2.3 One Categorical and One Numeric Variables

The categorical is usually served as a group identifier that splits the data into groups so the numerical variable can be compared between groups. This kind of graph is generally named side-by-side plot. You can choose a side-by-side boxplot or a side-by-side barplot. For example, visually compare the robust statistics between different species of iris.

#### 3.2.3.1 Side-by-side Boxplot

```
par(mfrow=c(2,1))

boxplot(Sepal.Length ~ Species,
        data=iris,
        main="A Side-by-Side Boxplot")

boxplot(Sepal.Length ~ Species,
        data=iris,
        horizontal = TRUE,
        main="A Horizontal Side-by-Side Boxplot")
```

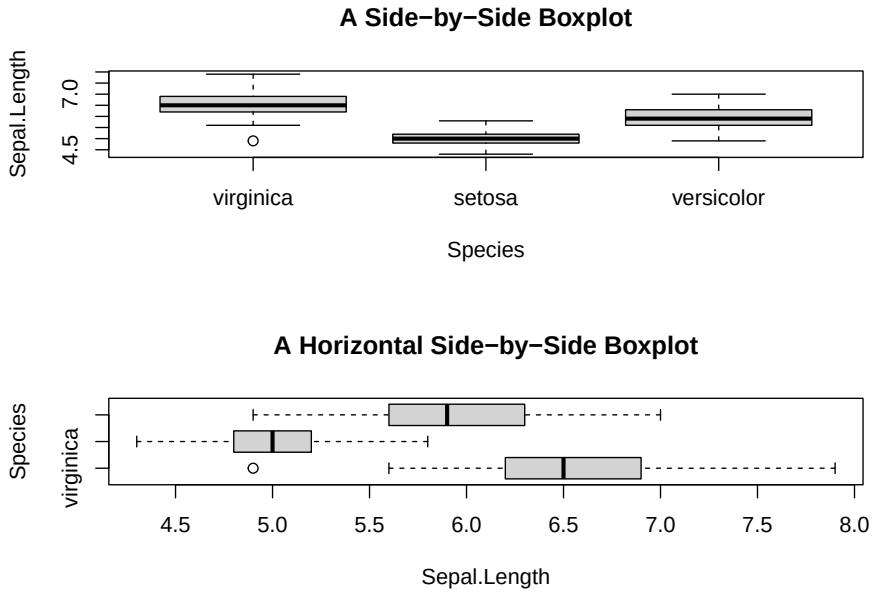



Figure 3.8: Side-by-side plot

`Sepal.Length ~ Species` is read as "Sepal length **by** species", the tilde symbol "`~`" means "by".

However, there is glitch in the horizontal boxplot. The tick y-labels collided. It can be fixed by adding `las=2` parameter to the `boxplot()` function.

```
boxplot(Sepal.Length ~ Species,
        data=iris,
        horizontal = TRUE,
        las = 2,
        main="Y-tick Labels Reorientated")
```

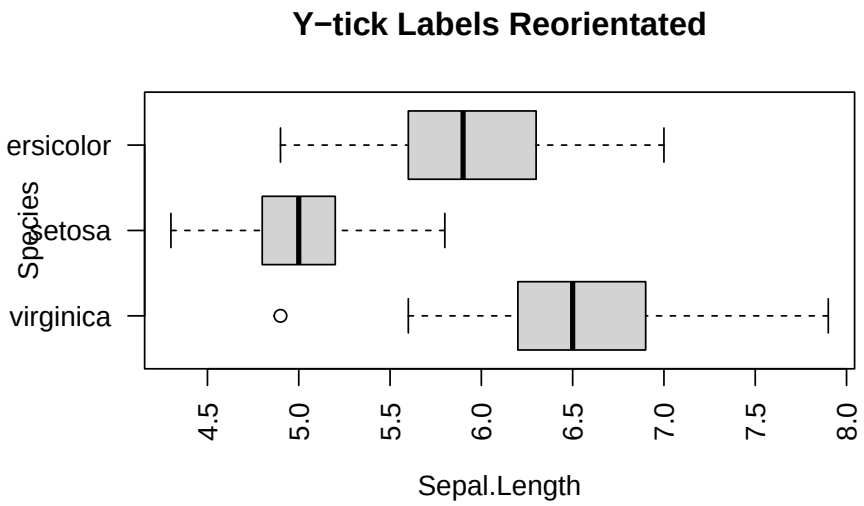



Figure 3.9: Side-by-side plot

#### 3.2.3.2 Side-by-side Barplot

Suppose you want to visually compare the average sepal length among different iris species. You will begin to feel the cumbersome of base R, and appreciate the power of `tidyverse`. Anyway, you will see how to make such a graph with a new function, `tapply()`, in two steps.

First, make a table of average sepal length by species. Apparently, the `table()` function may help. But it does not work for this case, as it only tallies the number of samples by species. Hence, you need to use a new function `tapply()`. Here's an example:

```
mean_data <- tapply(iris$Sepal.Length, iris$Species, FUN = mean)
mean_data
```

```
 virginica     setosa versicolor
     6.588      5.006      5.936
```

How does `tapply()` work? The "`t`" represents table. The function applies the `mean` function to the input data `iris$Sepal.Length` grouped by `iris$Species`. In other words, the `mean()` is applied to the sepal lengths for each species separately.

And then, pass the `mean_data` object to `barplot()`.

```
barplot(mean_data,
        xlab="Species",
        ylab="Average Sepal.Length",
        main="Side-by-side Barplot")
```

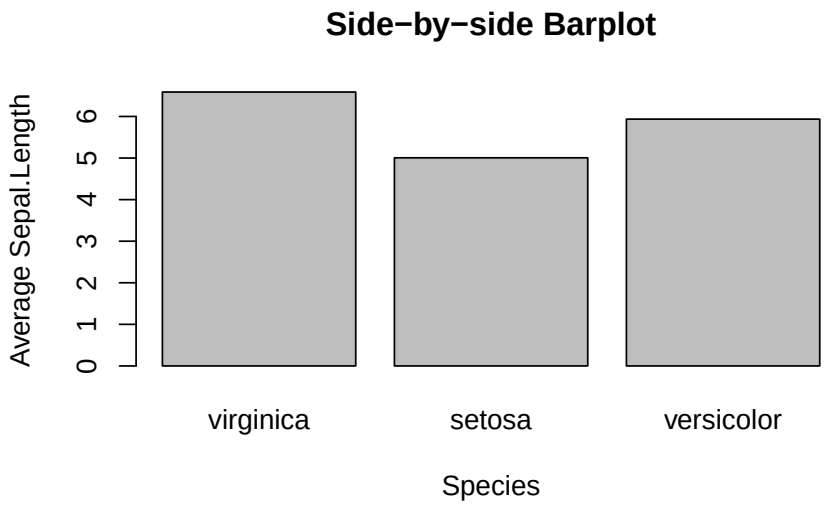



Figure 3.10: Side-by-side Barplot

#### 3.2.3.3 Side-by-side Barplot with Error Bar^*

*This topic is slightly advanced. It is much easier to use `ggplot2`. So, feel free to skip this and jump to @ref(ggplot-error-bar).

The error bar represents the standard error of the mean (SEM). Here is the formula:

$$\bar{x} \pm \frac{s}{\sqrt{n}}$$

, where $s$ and $n$ are the standard deviation and sample size, respectively. In the bar plot you are going to plot, the height of the bar is $\bar{x}$, and the error bars span above and below the bar's top by $\frac{s}{\sqrt{n}}$ units.

First, calculate the mean and standard of errors using the `tapply()` function.

```
mean_data <- tapply(iris$Sepal.Length, iris$Species, FUN = mean)
mean_data
```

```
 virginica     setosa versicolor
     6.588      5.006      5.936
```

```
sem <- function(x) {sd(x)/sqrt(length(x))}
sem_data <- tapply(iris$Sepal.Length, iris$Species, FUN = sem)
sem_data
```

```
 virginica     setosa versicolor
0.08992695 0.04984957 0.07299762
```

First, create the bar plot and store the height of bars in the object `bar_midpoints`. Use the `arrows()` function create the error bars, where each error bar consists of the upper whisker $(x_1, y_1)$ and lower whisker $(x_0, y_0)$.

```
bar_midpoints <- barplot(mean_data,
                  xlab="Species",
                  ylab="Average Sepal.Length",
                  main="Side-by-side Barplot")

arrows(x0 = bar_midpoints,
       y0 = mean_data - sem_data, # Start of the bar
       x1 = bar_midpoints,
       y1 = mean_data + sem_data, # End of the bar
       angle = 90,                # Makes the ends of the bars flat
       code = 3,                  # Draws ticks at both ends
       length = 0.5, col='red')   # Sets the length of the ticks
```

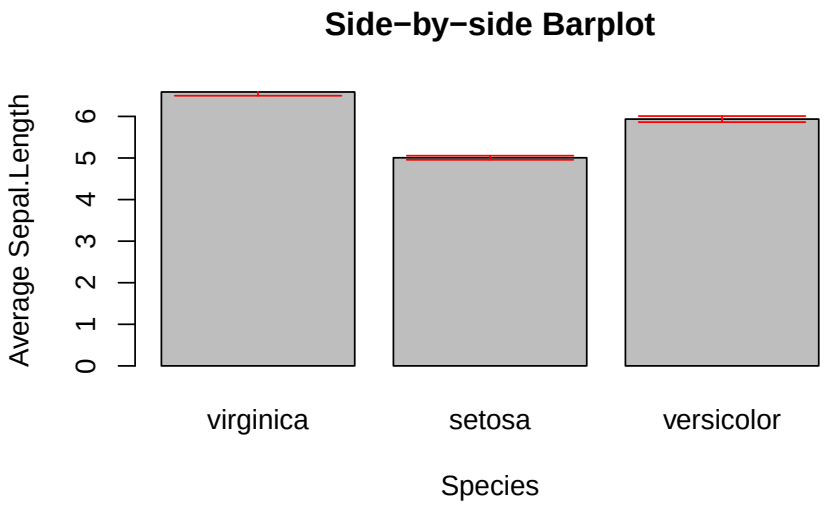



Figure 3.11: Side-by-side Barplot with Error Bars

Once again, if you found this confusing, see the `ggplot` version of error bars (Section @ref(ggplot-error-bar)).

#### 3.2.3.4 Line Graph with Error Bar

A line graph is an alternative to a bar plot that is good at showing a trend. Here you see how to create a line graph overlaid with solid dots.

```
# Default line plot
plot(mean_data,
      type='b',
      pch=19,
      xlab="Species",
      ylab="Average Sepal.Length",
      main="Line Plot")
```

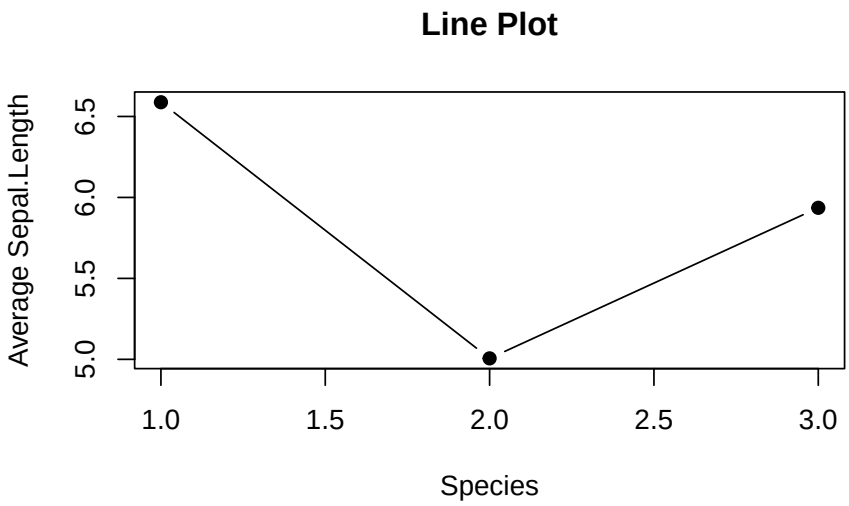



Figure 3.12: Line plot

- Use `pch=19` to specify the shape of the data points. `pch` stands for **p**lotting **ch**aracter. The value `19` denotes a solid dot (●).

- To join the dots together in a line plot, specify `type='b'`, the `b` means to plot **both** dots and line. More options of `type` can be found in the `help(plot)` documentation.

But, there is a glitch in the x-tick.The x-tick should show the actual species names instead of number. It can be fixed in two steps: 1. Suppress the default x-tick labeling feature of the `plot()` function by setting the parameter `xaxt="n"`. And then, paste the customized x-tick back by the `axis()` function. The species names can be obtained by `names(mean_data)`.

```
# Fix x-tick labels
plot(mean_data,
      type='b',
      pch=19,
      xlab="Species",
      ylab="Average Sepal.Length",
      main="Change x-tick Labels",
      xaxt="n")
axis(side=1, at=1:length(mean_data), labels=names(mean_data))
```

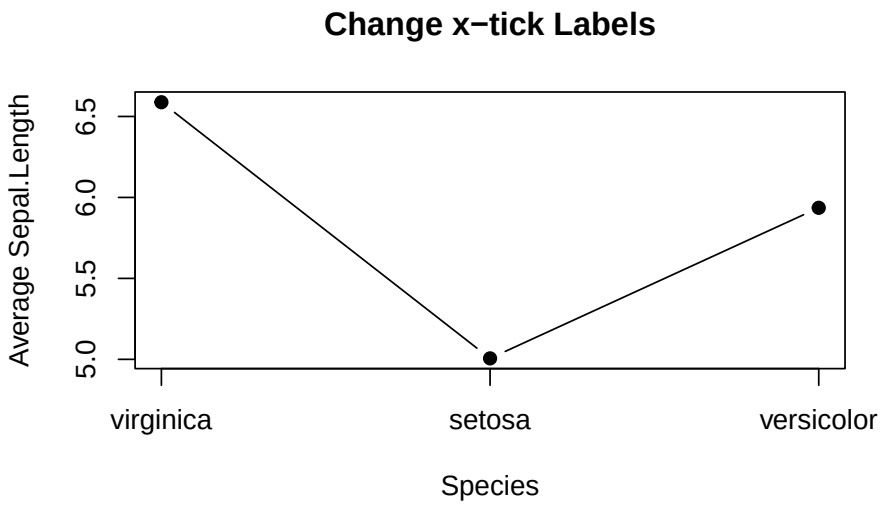



Figure 3.13: Line plot with proper x-tick labels

Here's how to add error bars to the line plot. It is similar to the barplot above, except that `x0` and `x1` should take the values `1,2,3`.

```
# Add Error Bar
plot(mean_data,
      type='b',
      pch=19,
      xlab="Species",
      ylab="Average Sepal.Length",
      main="Change x-tick Labels",
      xaxt="n")
axis(side=1, at=1:length(mean_data), labels=names(mean_data))

arrows(x0 = seq(length(mean_data)),
       y0 = mean_data - sem_data, # Start of the bar
       x1 = seq(length(mean_data)),
       y1 = mean_data + sem_data, # End of the bar
       angle = 90,                # Makes the ends of the bars flat
       code = 3,                  # Draws ticks at both ends
       length = 0.05, col='red')   # Sets the length of the ticks
```

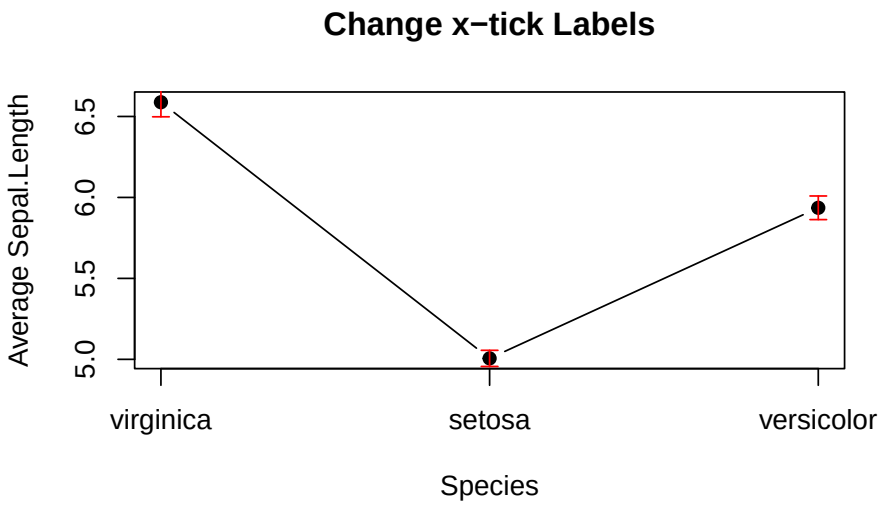



Figure 3.14: Line plot with error bars

#### 3.2.3.5 Overlapping Histogram^*

/* The code for creating this plot is a bit convoluted. It is easier to make it using `ggplot2`. Feel free to skip to @ref(ggplot-overlapping-histogram).

Suppose you are interested in comparing the distribution of sepal lengths among the three iris species. Here are the steps:

1. Split the data by individual species.
2. Calculate the range for the x- and y-axis so as to ensure the histograms are in the same scale.
3. Create a histogram for a data subset from step 1, which becomes the template of the final graph. Additionally, color is used to distinguish one species from the other.
4. Overlay the histograms for other data subsets on top of the template.

Let's do it step by step below:

```
# Step 1
setosa <- subset(iris, Species == "setosa")
versicolor <- subset(iris, Species == "versicolor")
virginica <- subset(iris, Species == "virginica")
```

```
# Step 2
x_range <- range(iris$Sepal.Length)
y_range <- c(0, 25) # 25 is arbitrary.
```

Below is the code for plotting the template histogram, and how additional histograms are added to it. To distinguish species, different color is attributed to different species. It is done by `col = rgb(...)`. The `rgb()` function stands for **r**ed, **g**reen, and **b**lue. It takes four parameters: the intensity of red, green, blue, and transparency. All are in the range of 0 and 1. E.g., `rgb(1, 0, 0, 0.5)` means 100% red, no green and blue at all, and transparency is 50%.

```
# Step 3
hist(setosa$Sepal.Length,
     main = "Overlapping Histogram of Sepal Length",
     xlab = "Sepal Length",
     col = rgb(1, 0, 0, 0.5), # Red with 50% transparency
     breaks = seq(4,9,by=0.25),
     xlim = x_range,
     ylim = y_range)

# Step 4
hist(versicolor$Sepal.Length, col=rgb(0, 1, 0, 0.5), add = TRUE) # green with 50% trans
hist(virginica$Sepal.Length, col=rgb(0, 0, 1, 0.5), add = TRUE)  # blue with 50% transp
```

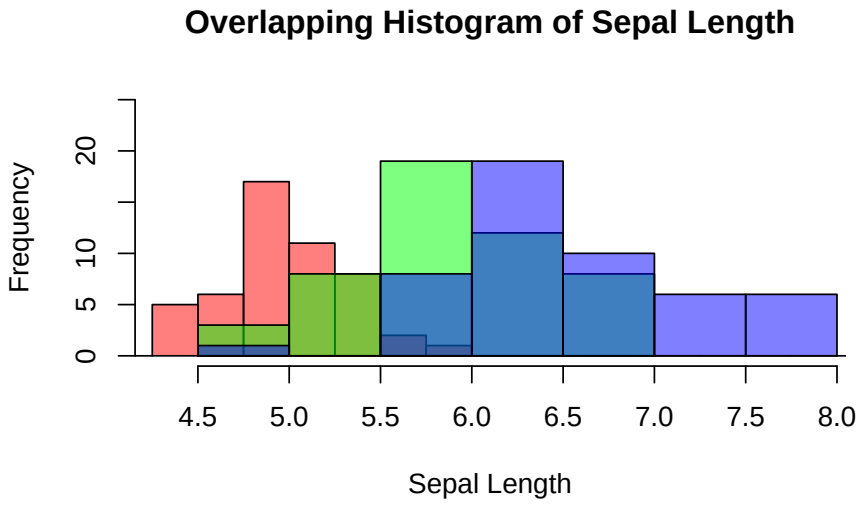



Figure 3.15: Overlapping histogram

### 3.2.4 Contingency Table

If a dataset comprises two categorical variables. A two-dimensional contingency table, or a 2D table, depicts the number of samples present in different combinations of categories of the two categorical variables. The distribution of the counts can reveal the dependency of the two categorical variables. Let's use another built-in dataset `penguins` for illustration. It contains two categorical variables: `species` and `island`. Suppose you are interested to explore how geographical location affects the distribution of penguins.

Create a 2D table as below:

```
mytab <- with(penguins, table(species, island))
mytab
```

```
           island
species     Biscoe Dream Torgersen
  Adelie        44    56        52
  Chinstrap      0    68         0
  Gentoo       124     0         0
```

Obviously, `species` and `island` are not independent, as you can see not all three species can be found in each island. If `species` and `island` are independent, the counts in the contingency table should be below:

```
indep_tab
```

```
           island
species     Biscoe Dream Torgersen
  Adelie        74    55        23
  Chinstrap     33    25        10
  Gentoo        61    45        19
```

Two popular graphs are used to visualize counts from a contingency table: a side-by-side barplot, and mosaic plot.

```
barplot(mytab,
        beside=T,
        xlab = "Island",
        ylab = "Count",
        main = "A 2D Table",
        col = c("skyblue", "lightgreen", "lightcoral"),
        legend.text = rownames(mytab))
```

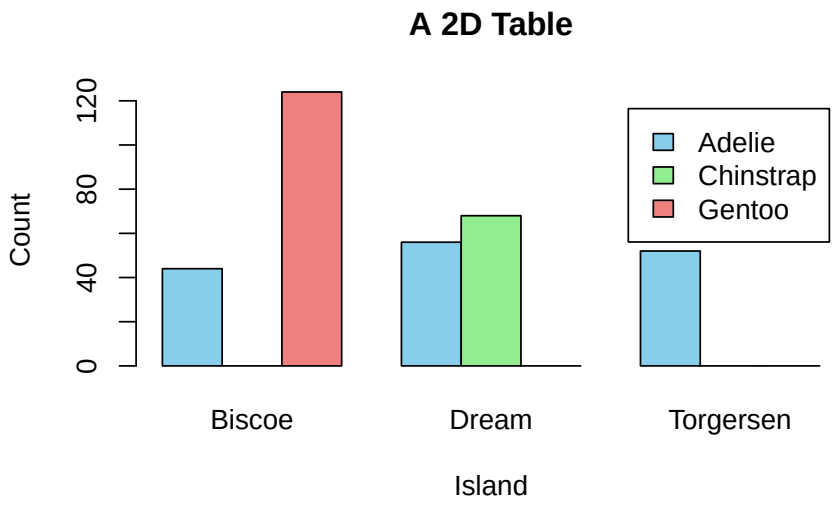



Figure 3.16: Visualization of a 2D Contingency Table. A. A side-by-side barplot. B. Mosaic plot.

```
mosaicplot(mytab, main = "Mosaic Plot of Two Dependent Variables")
```

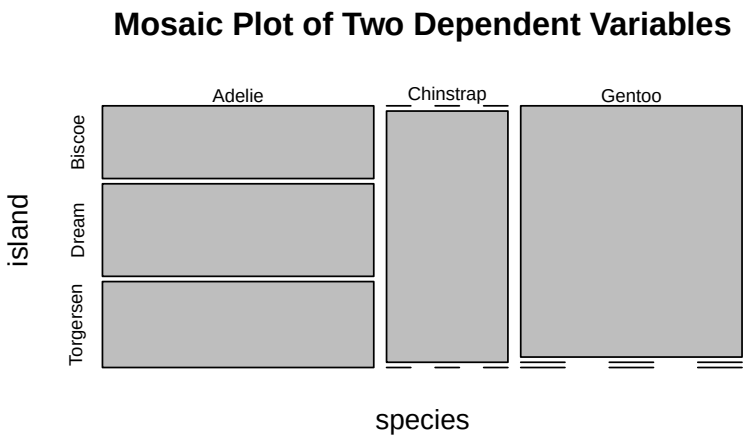



Figure 3.17: Visualization of a 2D Contingency Table. A. A side-by-side barplot. B. Mosaic plot.

```
mosaicplot(indep_tab, main = "Mosaic Plot of Two Independent Variables")
```

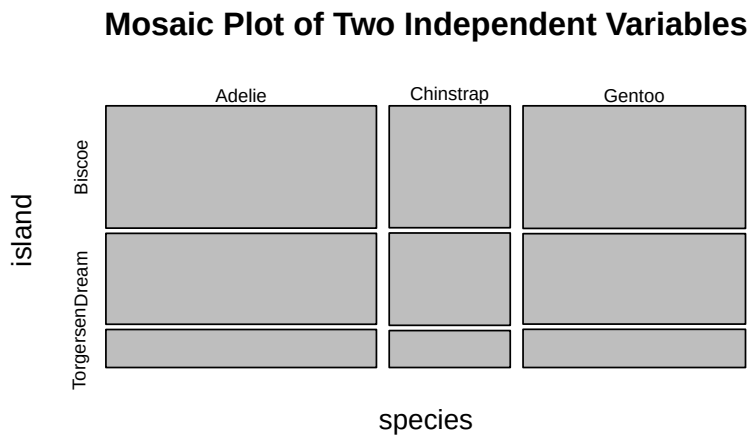



Figure 3.18: Visualization of a 2D Contingency Table. A. A side-by-side barplot. B. Mosaic plot.

### 3.2.5 Two Numeric Variables

If you want to gain a gut feeling about the relationship between two numeric variables, a scatter plot is the best choice. Note that the relationship can be manifold, ranging from linear to cyclical. So, visualizing how the data points are distributed in a 2D plan is a quick and easy way in gaining intuition of the two variables. Importantly, it should be done before performing any statistical analysis, e.g., linear regression. Here is how to create a scatter plot that shows the relationship between sepal width and sepal length. Note that each point $(x_i, y_i)$ is a pair of values from one sample.

```
par(mfrow=c(1,2))

# Top left
plot(x=iris$Sepal.Width,
     y=iris$Sepal.Length,
     xlab = "Sepal Width",
     ylab = "Sepal Length",
     main = "Default")

# Top right
plot(x=iris$Sepal.Width,
```

```
     y=iris$Sepal.Length,
     pch=3,
     xlab = "Sepal Width",
     ylab = "Sepal Length",
     main = "Change the Point Style")
```

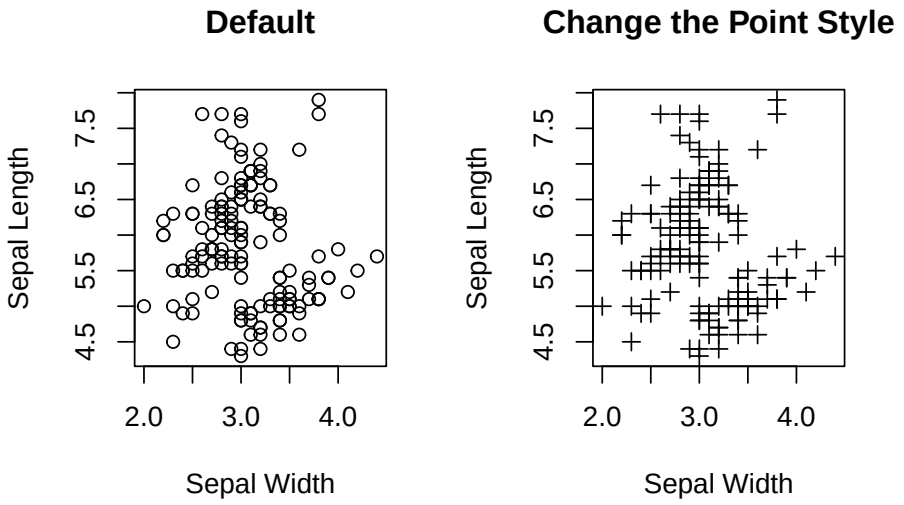



Figure 3.19: Visualizing the relationship between two numeric variables.

```
plot(x=iris$Sepal.Width,
     y=iris$Sepal.Length,
     pch=3,
     xlab = "Sepal Width",
     ylab = "Sepal Length",
     main = "Segregate Data Points by Color (Species)",
     col = c("skyblue", "lightgreen", "lightcoral")[as.numeric(iris$Species)])

legend("topright",
       legend = levels(iris$Species),
       col = c("skyblue", "lightgreen", "lightcoral"),
       pch = 3,
       title = "Species")
```

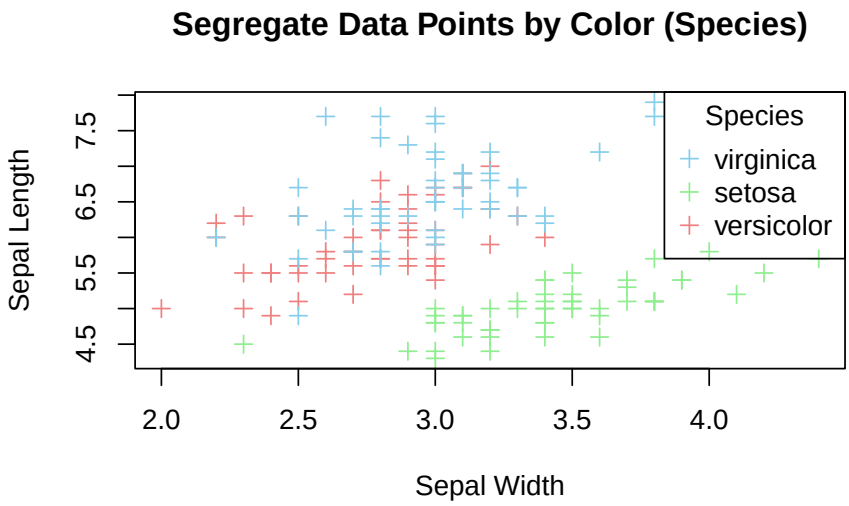



Figure 3.20: Color data points by species.

If your aim is to explore linear relationship between the two variables, it can be emphasized by plotting a regression line for the two variables on top of the data points.

```
lm_model <- lm(Sepal.Length ~ Sepal.Width, data=iris)

# intercept
intercept <- lm_model$coefficients[1]

# slope
slope = lm_model$coefficients[2]

print(paste("Slope =", slope, "Intercept =", intercept))
```

```
[1] "Slope = -0.2233610611299 Intercept = 6.52622255089448"
```

```
plot(x=iris$Sepal.Width,
     y=iris$Sepal.Length,
     pch=3,
     xlab = "Sepal Width",
     ylab = "Sepal Length",
     main = "Add a Regression Line",
     col = c("skyblue", "lightgreen", "lightcoral")[as.numeric(iris$Species)])
```

```
legend("topright",
       legend = levels(iris$Species),
       col = c("skyblue", "lightgreen", "lightcoral"),
       pch = 3,
       title = "Species")

# Add a line
abline(a=intercept, b=slope, col='red')
```

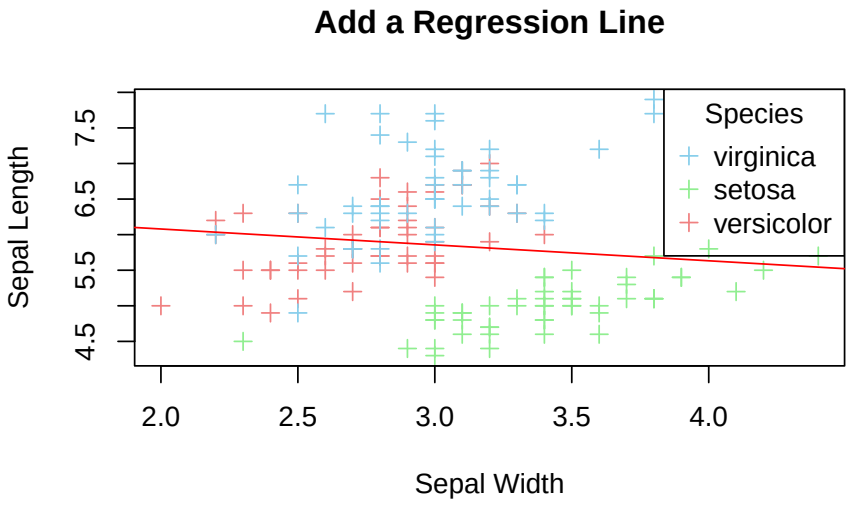



Figure 3.21: Add a regression line

### 3.2.6 Multi-panel Scatter Plot

A multi-panel scatter plot offers a glimpse of the relationship between all possible pairs of numeric variables. Suppose, you are interested in the pairwise relationship between all numeric variables in `iris`. It can easily be done using the `pairs()` plot function.

```
pairs(iris[,1:4], main="A Multi-Panel Scatter Plot")
```

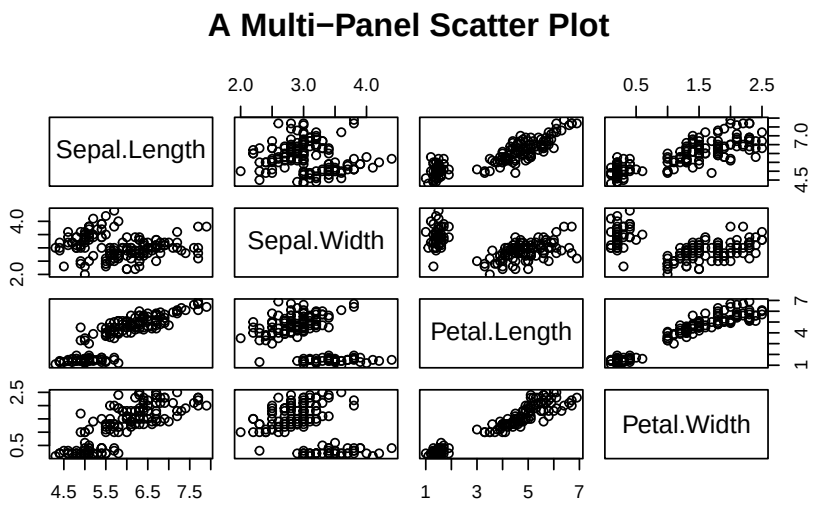



Figure 3.22: A Multi-panel Scatter Plot

Here, it concludes our journey of data visualization using base R functions. As you can see, some plots can be made easily using base R (e.g., multi-panel scatter plot), while others are complicated (e.g., barplot with error bars). To compare the pros and cons of the two graphical systems, let's continue our journey to `ggplot`.

## 3.3 ggplot2

`ggplot2` is bundled in `tidyverse`. So, you have to activate `tidyverse` before using `ggplot2`. Here, the focus is on using `ggplot2`to plot the graphs that were created using base R above. We will not repeat the discussion of strengths and weaknesses of each graph in this section. Readers should refer to the corresponding section above for the characteristics and functions of the graphs.

To use `ggplot2`, load it into R.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
```

```
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ---------------------------------------- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become er
```

### 3.3.1 Grammer of ggplot2

The two 'g's of ggplot2 stand for **Grammar of Graphics**. The graphical widgets are arranged in layers that provides the flexibility for decorating a graph by adding layers to it. The **data** layer forms the foundation of a graph. The **Aesthetics** defines the association between data variable(s) and the axes. **Geometries** are the types of graphs, e.g., line plot, histogram, etc. **Facets** split a graph into multi-panels. **Statistics** defines how to aggregate data, e.g., in proportion or take the face value in a barplot. **Coordinates** supports an X-Y Cartesian coordinate system, and a polar coordinate system. (See the Pie Chart section below) **Themes** offers different color schemes, font type, font size, etc.

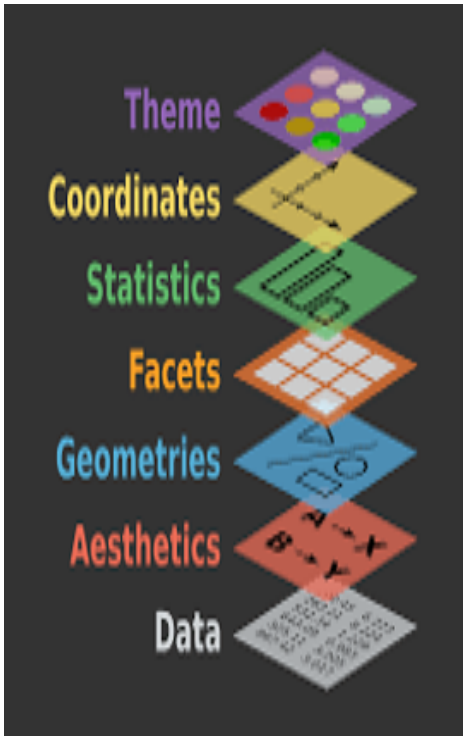



Figure 3.23: Anatomy of ggplot2

### 3.3.2 One Numeric Variable

You can plot a histogram of the numeric variable to see its distribution, such as the spread, the peak, symmetry, and flatness. If your concern is robust statistics, a boxplot will do the job.

#### 3.3.2.1 Histogram

Let's redraw the histogram in Figure (@ref(fig:Fig3-1))) using `ggplot2`. The first line is the data layer that specifies the data source, and it must be a data frame (or tibble), e.g., `iris`. The aesthetic option designates the x-axis for `Sepal.Width`. The second line is the geometry of the graph. For a histogram, the function name is `geom_histogram()`. Here is the bare minimum of a `ggplot2` histogram:

```
# Default
ggplot(data=iris, aes(x=Sepal.Width)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

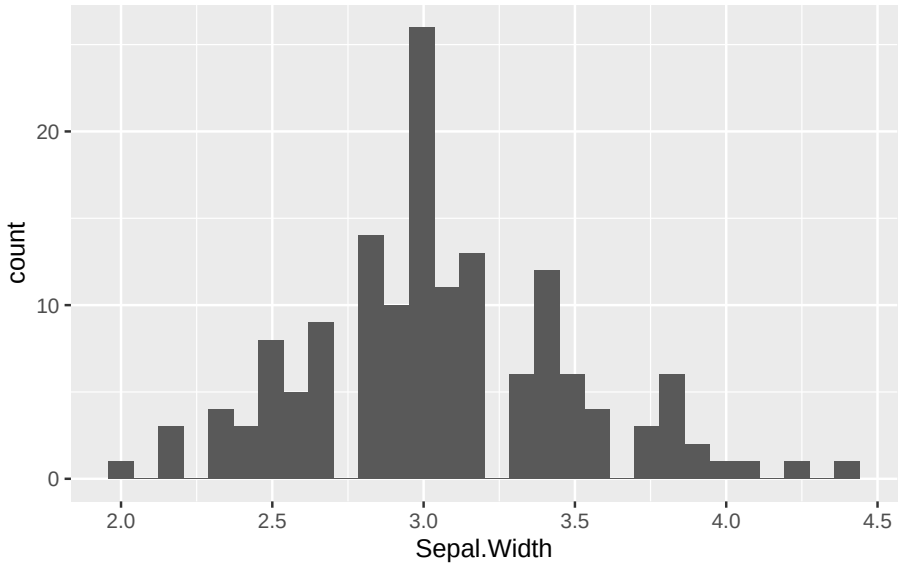



Figure 3.24: `ggplot2` Histogram.

It is always a good idea to give the graph a title at the top so people what the key message is. Equally important is the x- and y-labels. This task can be done

by the `labs()` function. By default, the title is left-justified. To center it, see the last line and the comment above it.

```
# Overwrite default x-/y-label and title
ggplot(data=iris, aes(x=Sepal.Width)) +
  geom_histogram() +
  labs(x="Sepal Width", y="Count", title="Histogram of Sepal Width") +
  # hjust = 0.5 means the horizontal adjustment is the midpoint.
  theme(plot.title = element_text(hjust = 0.5))
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

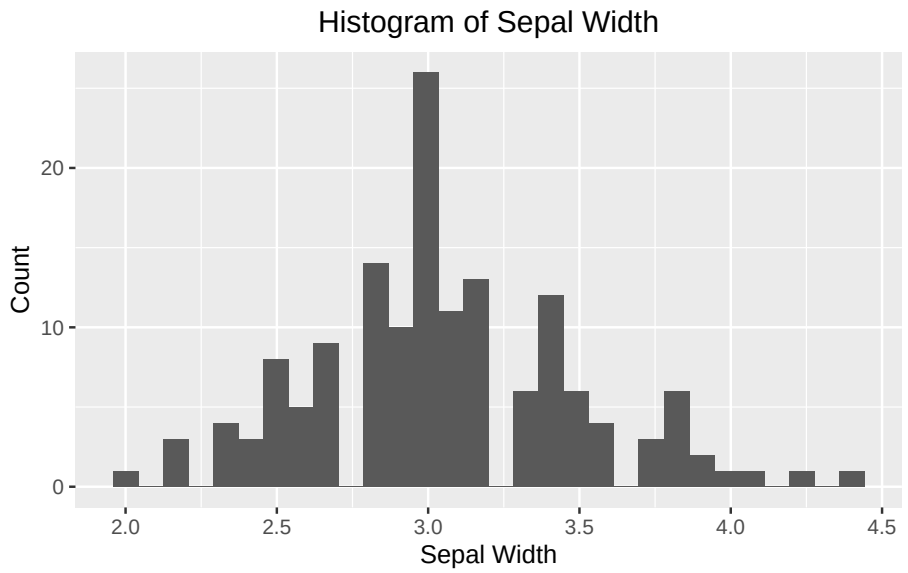



Figure 3.25: `ggplot2` Histogram - specify your own labels and title.

Change the fill color and border color. It can be done by providing additional parameters to the `geom_histogram()`.

```
# fill = filled color
# color = border color
ggplot(data=iris, aes(x=Sepal.Width)) +
  geom_histogram(fill="skyblue", color = "white") +
  labs(x="Sepal Width", y="Count", title="Color Options") +
  # hjust = 0.5 means the horizontal adjustment is the midpoint.
  theme(plot.title = element_text(hjust = 0.5))
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

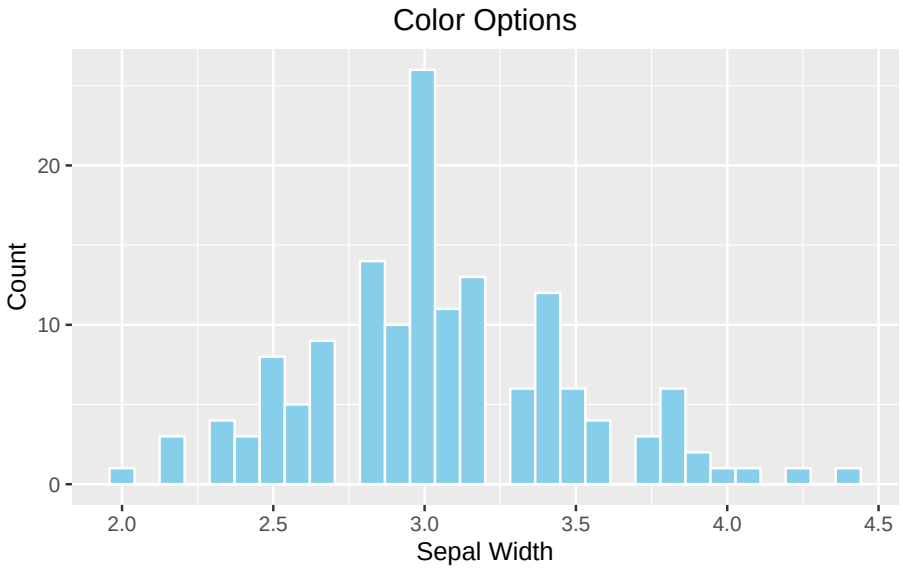



Figure 3.26: `ggplot2` Histogram - choose your own color

The bin width of the histogram is calculated automatically by `ggplot2` according to the range of x. To overwrite it by the `breaks` parameter.

```
# breaks
ggplot(data=iris, aes(x=Sepal.Width)) +
  geom_histogram(fill="skyblue", color = "white", breaks=seq(2,4.5,by=0.1)) +
  labs(x="Sepal Width", y="Count", title="`breaks` option") +
  # hjust = 0.5 means the horizontal adjustment is the midpoint.
  theme(plot.title = element_text(hjust = 0.5))
```

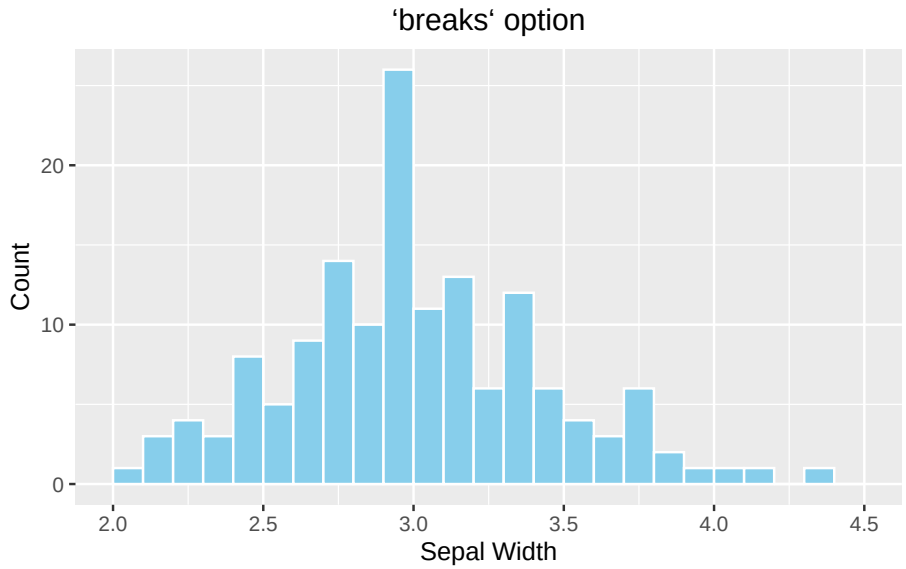



Figure 3.27: `ggplot2` Histogram - choose your own color

### Optional topic

As you might notice that each of the four histograms occupies the entire plotting area instead of displaying in a 2x2 grid as Figure (@ref(fig:Fig3-1))). It is because `ggolot2` does not work with `par(mfrow(2,2))`. To divide the plotting area into a grid that is compatible with `ggplot2`, use the `patchwork` package. Use the menu option **Tools** → Install Packages to install it the very first time.

Another new concept is that a graph can be stored in a variable so it can be displayed later in the grid. In the example below, the four histograms above are initially stored in variables `p1`, `p2`, `p3`, and `p4`. See the example below:

```
library(patchwork)

# Default
p1 <- ggplot(data=iris, aes(x=Sepal.Width)) +
        geom_histogram()

# With labels and title
p2 <- ggplot(data=iris, aes(x=Sepal.Width)) +
        geom_histogram() +
        labs(x="Sepal Width", y="Count", title="Histogram of Sepal Width") +
        # hjust = 0.5 means the horizontal adjustment is the midpoint.
        theme(plot.title = element_text(hjust = 0.5))
```

```
# fill = filled color
# colour = border color
p3 <- ggplot(data=iris, aes(x=Sepal.Width)) +
        geom_histogram(fill="skyblue", color = "white") +
        labs(x="Sepal Width", y="Count", title="Color Options") +
        # hjust = 0.5 means the horizontal adjustment is the midpoint.
        theme(plot.title = element_text(hjust = 0.5))

# Change the number of bins
p4 <- ggplot(data=iris, aes(x=Sepal.Width)) +
        geom_histogram(fill="skyblue", color = "white", breaks=seq(2,4.5,by=0.1)) +
        labs(x="Sepal Width", y="Count", title="`breaks` option") +
        # hjust = 0.5 means the horizontal adjustment is the midpoint.
        theme(plot.title = element_text(hjust = 0.5))

# Arrange them in a 2x2 grid
# '+' places plots next to each other, '/' stacks them
(p1 + p2) / (p3 + p4)
```

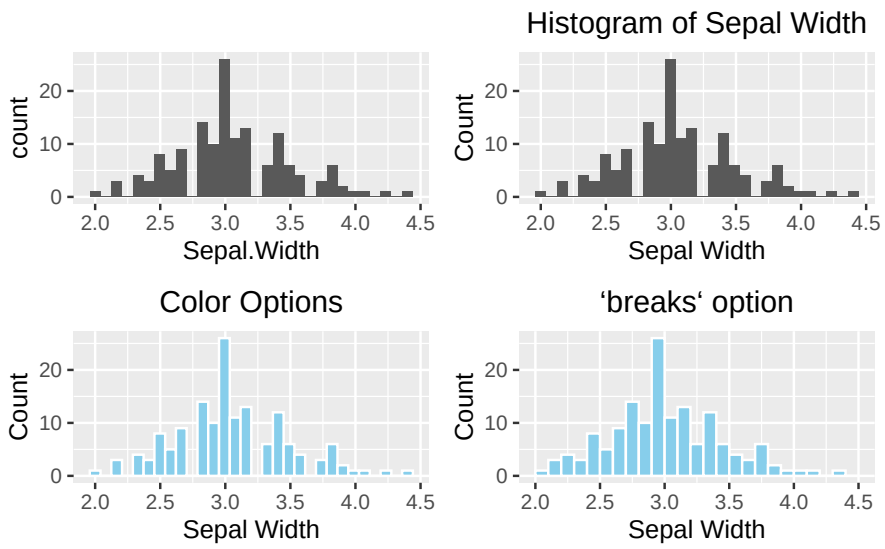



Figure 3.28: `ggplot2` Grid System by `patchwork`. Top left: Default (bare minimum). Top right: with labels and title. Bottom left: change the color. Bottom right: change the number of bins.

#### 3.3.2.2 Boxplot

For an anatomy of a boxplot, reference the Figure (@ref(fig:Fig3-3))). The geometry function of a boxplot is `geom_boxplot()`. Unlike a histogram, the numeric variable is mapped to the y-axis for a vertical boxplot. If you want a horizontal boxplot, map the numeric variable to the x-axis, and don't forget to specify the label for the x-axis.

```
library(patchwork)

# Default
p1 <- ggplot(data=iris, aes(y=Sepal.Width)) +
        geom_boxplot()

# Add y-label and title
p2 <- ggplot(data=iris, aes(y=Sepal.Width)) +
        geom_boxplot() +
        labs(y="Sepal Width", title = "Boxplot of Sepal Width") +
        theme(plot.title = element_text(hjust = 0.5))

# Change color
p3 <- ggplot(data=iris, aes(y=Sepal.Width)) +
        geom_boxplot(fill="skyblue", outlier.color = "red") +
        labs(y="Sepal Width", title = "Color options") +
        theme(plot.title = element_text(hjust = 0.5))

# Horizontal boxplot
p4 <- ggplot(data=iris, aes(x=Sepal.Width)) +
        geom_boxplot() +
        labs(x="Sepal Width", title = "Horizontal Boxplot") +
        theme(plot.title = element_text(hjust = 0.5))

(p1 + p2)/(p3 + p4)
```

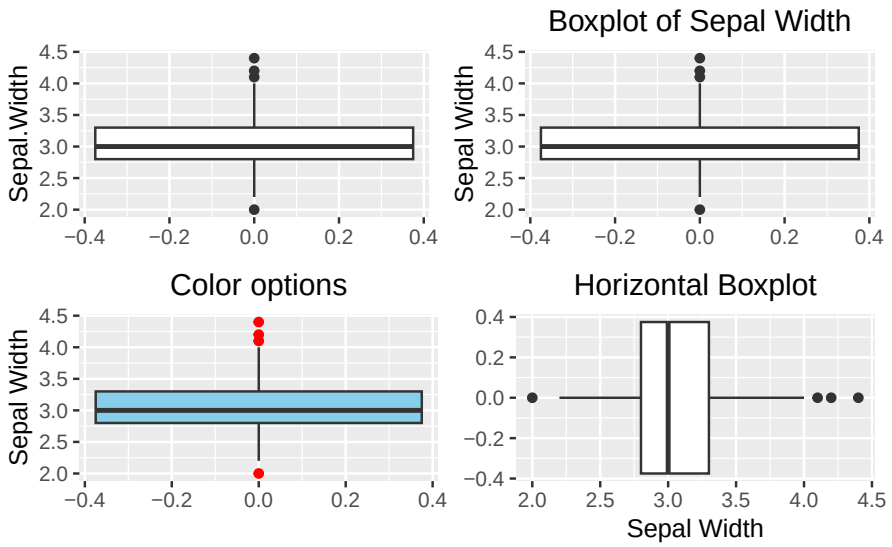



Figure 3.29: Boxplot using `ggplot2`. Top left: default (bare minimum). Top right: with y-label and title. Bottom left: change color. Bottom right: horizontal boxplot.

### 3.3.3 One Categorical Variable

#### 3.3.3.1 Barplot

Here we will show how to make a barplot and a piechart using `ggplot2`.

```
library(patchwork)

# Default
p1 <- ggplot(data=iris, aes(x=Species)) +
        geom_bar()

# With labels and title
p2 <- ggplot(data=iris, aes(x=Species)) +
        geom_bar() +
        labs(x="Species", y="Count", title = "A Barplot of Species") +
        theme(plot.title = element_text(hjust = 0.5))

# Color option
p3 <- ggplot(data=iris, aes(x=Species)) +
        geom_bar(fill="skyblue", color="blue") +
```

```
        labs(x="Species", y="Count", title = "A Barplot of Species") +
        theme(plot.title = element_text(hjust = 0.5))

# Horizontal barplot
p4 <- ggplot(data=iris, aes(y=Species)) +
        geom_bar(fill="skyblue", color="blue") +
        labs(y="Species", x="Count", title = "A Horizontal Barplot of Species") +
        theme(plot.title = element_text(hjust = 0.5))

(p1 + p2)/(p3 + p4)
```

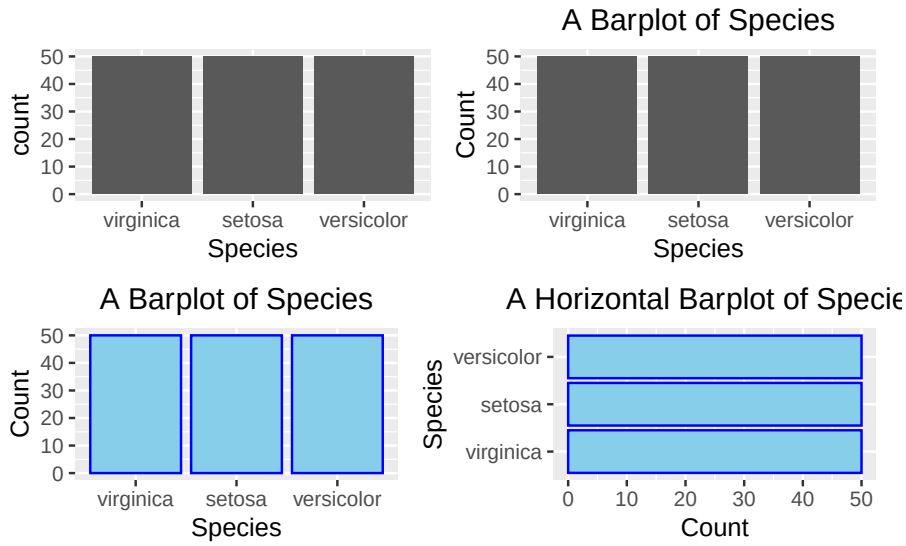



Figure 3.30: Barplot using `ggplot2`. Top left: default (bare minimum).

#### 3.3.3.2 Pie Chart

Although `ggplots` is portrayed as an improved method to compose graphs based upon its grammatical architecture, it does not always make plotting as simple as you might think, like the histograms and barplots illustrated above. A pie chart is one of those graphs in which `geom_pie()` does not exist! Plotting it is not as straightforward as the base R `pie()` function, as shown in Figure (@ref(fig:Fig3-7))). Despite that, it is worth to see how a pie chart can be plotted using foundational building-block functions. To give you a heads-up, `ggplot2` views a pie chart as a twisted barplot. There is an internal logic to justify such a view but we will not discuss it in detail.

Suppose we want to create a pie chart like the one below:

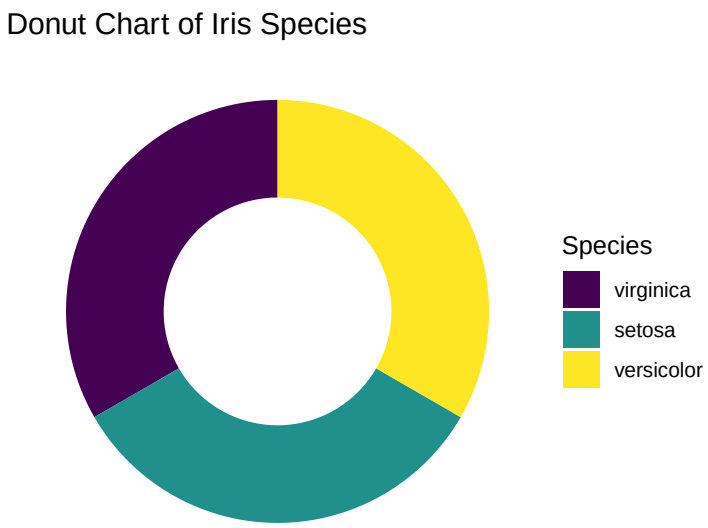



Figure 3.31: Pie chart or a donut chart

Here we will walk through each of the intermediate steps until getting the final pie chart.

Let's begin with a stacked barplot, where the bars are stacked up into a single column, and the fill-color is an attribute to distinguish different iris species (`fill=Species`). The stacked bar is placed at the position 2 units on the right from the origin (`aes(x=2)`), but the exact position does not matter.

```
ggplot(data=iris, aes(x=2, fill=Species)) +
  geom_bar()
```

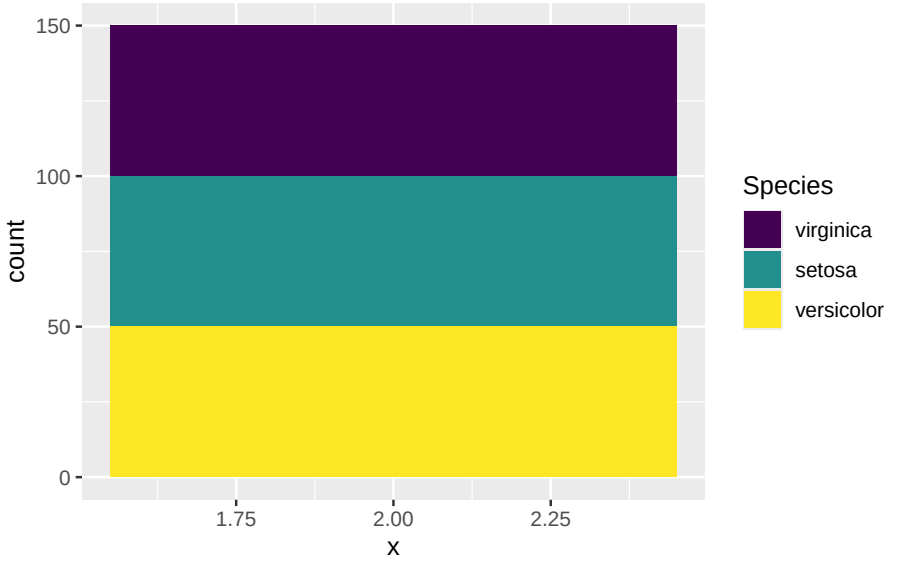



Figure 3.32: A stacked barplot

Now, use your imagination to picture that the stacked bar is bent into a ring. Does it look like the pie chart in Figure (@ref(fig:Fig3-30)))? To way to "bend" it in `ggplot2` is to change the x-y Cartesian coordinate system to the polar (circular) coordinate system in which a point is characterized by two parameters: the distance from the origin ($r$), and the angle ($\theta$) from the horizontal **polar axis** in an counter-clockwise direction.

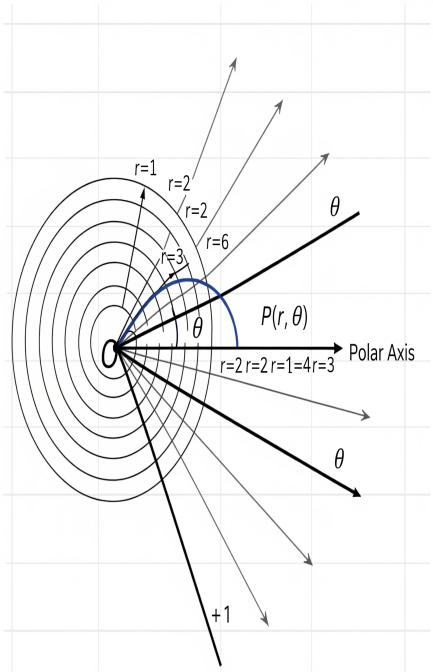



Figure 3.33: Polar Coordinate. (generated by Gemini)

By adding this line `coord_polar(theta="y", start = 0)` in the code below, it transforms the height of the bar for each species into an angle ($\theta$) in the polar coordinate, where the zero position is the 12 o'clock (`start = 0`). In this example, one third of the samples is `setosa`. As such, the angle $\theta$ is $360^\circ \times \frac{1}{3} = 120^\circ$. If the proportion is a quarter, the angle $\theta = 360^\circ \times \frac{1}{4} = 90^\circ$.

```
ggplot(data=iris, aes(x="", fill=Species)) +
  geom_bar() +
  coord_polar(theta="y", start = 0)
```

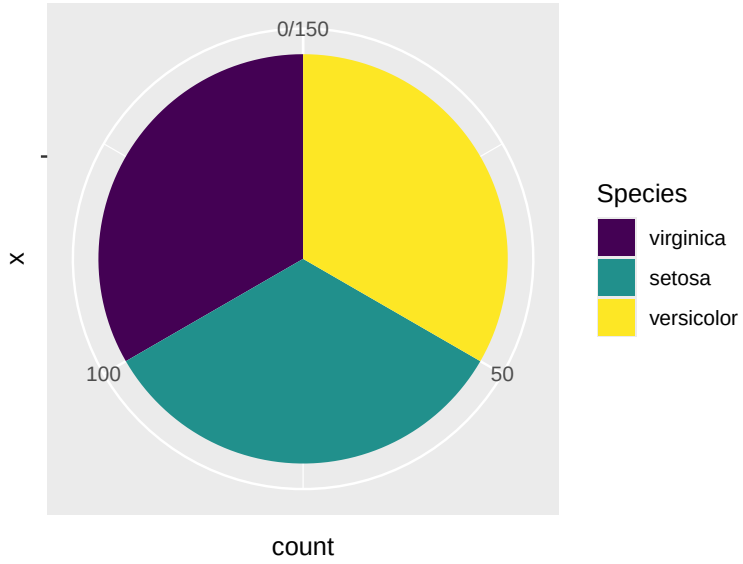



Figure 3.34: A Piechart by changing to the polar coordinate system.

The final step is to open a hole in the pie chart if you like, for aesthetic reasons. But this modification is optional.

```
ggplot(iris, aes(x = 2, fill = Species)) +
  geom_bar(width = 1, stat = "count") +
  coord_polar(theta = "y", start = 0) +
  # This is the key part that creates the hole
  xlim(0.5, 2.5) +
  labs(title = "Donut Chart of Iris Species")
```

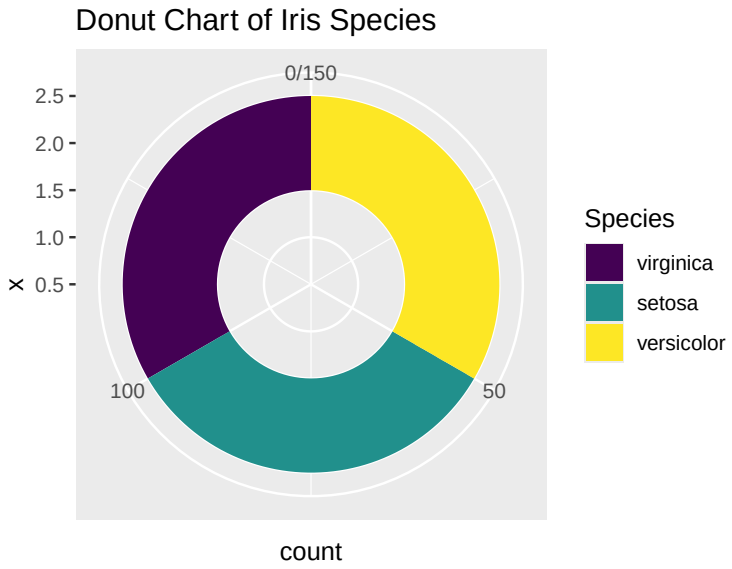



Figure 3.35: Piechart using `ggplot2`

If you prefer to have a clean background, add the line `theme_void()`. Finally, here's the desired pie chart. What a journey!

```
ggplot(iris, aes(x = 2, fill = Species)) +
  geom_bar(width = 1, stat = "count") +
  coord_polar(theta = "y", start = 0) +
  # This is the key part that creates the hole
  xlim(0.5, 2.5) +
  theme_void() +
  labs(title = "Donut Chart of Iris Species")
```

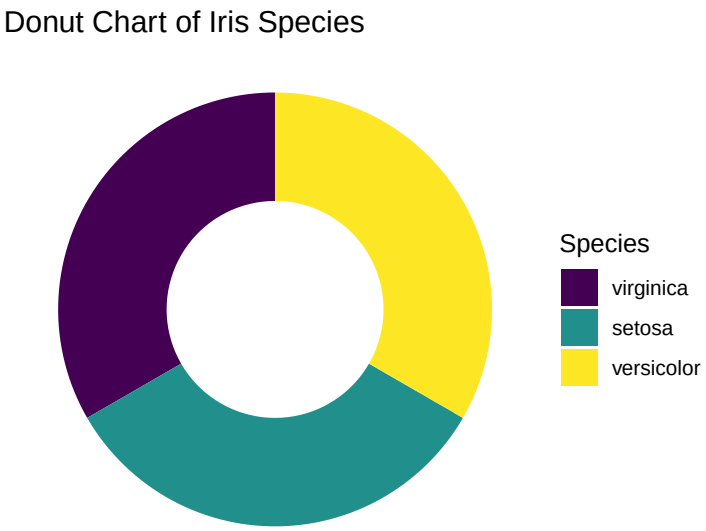



Figure 3.36: Piechart using `ggplot2` without no background.

### 3.3.4 One Categorical and One Numeric Variables

#### 3.3.4.1 Side-by-side Boxplot

Here you want to compare the robust statistics among different species.

```
library(patchwork)

p1 <- ggplot(data=iris, aes(x=Species, y=Sepal.Length)) +
        geom_boxplot() +
        labs(title = "A Side-by-Side Boxplot") +
        theme(plot.title = element_text(hjust = 0.5))

p2 <- ggplot(data=iris, aes(x=Sepal.Length, y=Species)) +
        geom_boxplot() +
        labs(title = "A Horizontal Side-by-Side Boxplot") +
        theme(plot.title = element_text(hjust = 0.5))

p1/p2
```

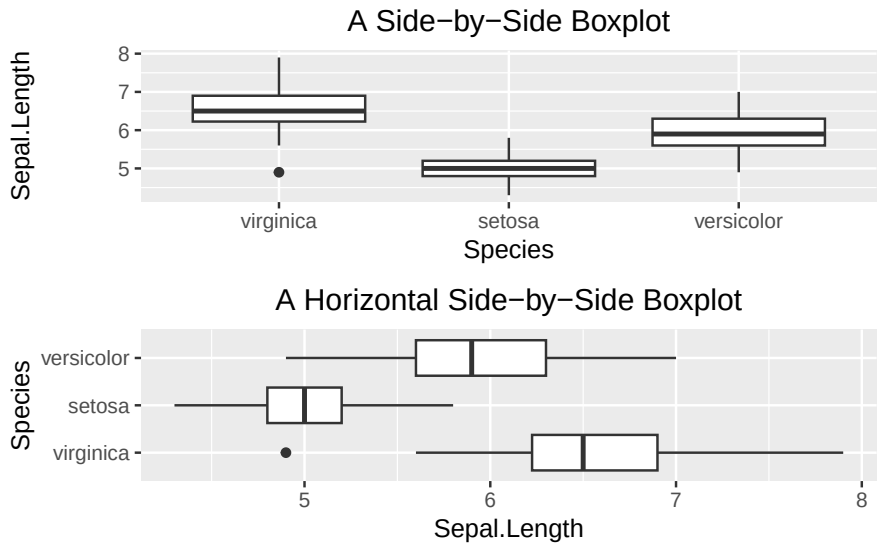



Figure 3.37: Side-by-side boxplot

#### 3.3.4.2 Side-by-side Barplot

Here, you want to compare the average `Sepal.Length` among iris species. As discussed above, you calculate the averages and store them in a separate table.

```
iris_summary1 <- iris %>%
                  group_by(Species) %>%
                  summarize(mean_sepal_length = mean(Sepal.Length))
iris_summary1
```

```
# A tibble: 3 x 2
  Species    mean_sepal_length
  <ord>                  <dbl>
1 virginica               6.59
2 setosa                  5.01
3 versicolor              5.94
```

And then, you can plot the summary table. The geometry function is `geom_bar(stat = "identity")`. The `stat = "identity"` parameter tells the `geom_bar()` function to set up the height of the bars according to the face value of `mean_sepal_length`.

```
ggplot(data=iris_summary1, aes(x=Species, y=mean_sepal_length)) +
  geom_bar(stat = "identity") +
  labs(x="Species", y="Average Sepal Length", title = "Side-by-side Barplot") +
  theme(plot.title = element_text(hjust = 0.5))
```

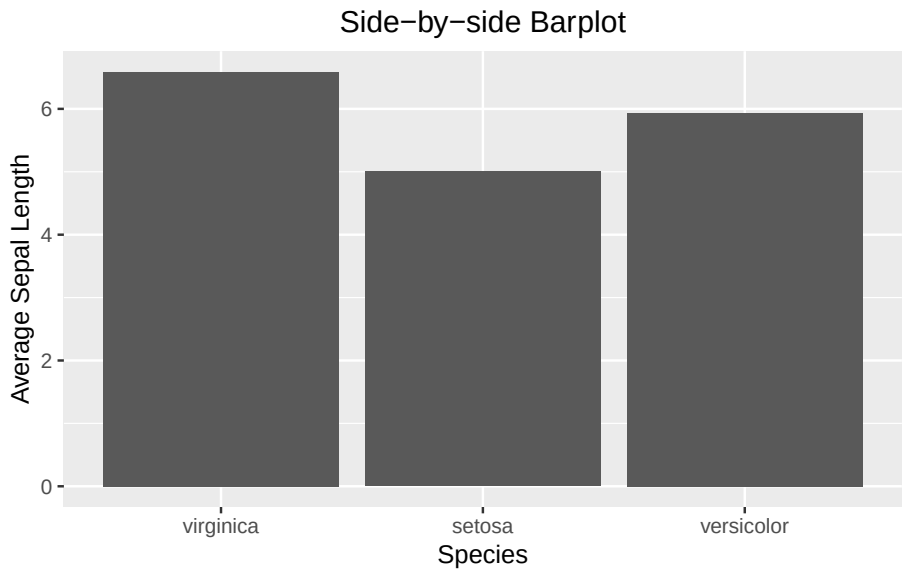



Figure 3.38: Side-by-side barplot

#### 3.3.4.3 Side-by-side Barplot with Error Bar

The calculation of the error bar (i.e., the standard error `se`) requires standard deviation and sample size. Once `se` is calculated, it can be used to calculate the minimum (lower) and maximum (upper) of the error bar.

```
iris_summary2 <- iris %>%
                  group_by(Species) %>%
                  summarize(mean_sepal_length = mean(Sepal.Length),
                            N=n(),
                            se=sd(Sepal.Length)/sqrt(N)) %>%
                  # add the y minimum and y maximum columns
                  mutate(ymin = mean_sepal_length - se,
                         ymax = mean_sepal_length + se)
iris_summary2
```

```
# A tibble: 3 x 6
```

```
  Species    mean_sepal_length     N     se  ymin  ymax
  <ord>                  <dbl> <int>  <dbl> <dbl> <dbl>
1 virginica               6.59    50 0.0899  6.50  6.68
2 setosa                  5.01    50 0.0498  4.96  5.06
3 versicolor              5.94    50 0.0730  5.86  6.01
```

```
ggplot(data=iris_summary2, aes(x=Species, y=mean_sepal_length)) +
  geom_bar(stat = "identity") +
  geom_errorbar(aes(ymin = ymin, ymax = ymax), width = 0.2, color='red') +
  labs(x="Species", y="Average Sepal Length", title = "Side-by-side Barplot") +
  theme(plot.title = element_text(hjust = 0.5))
```

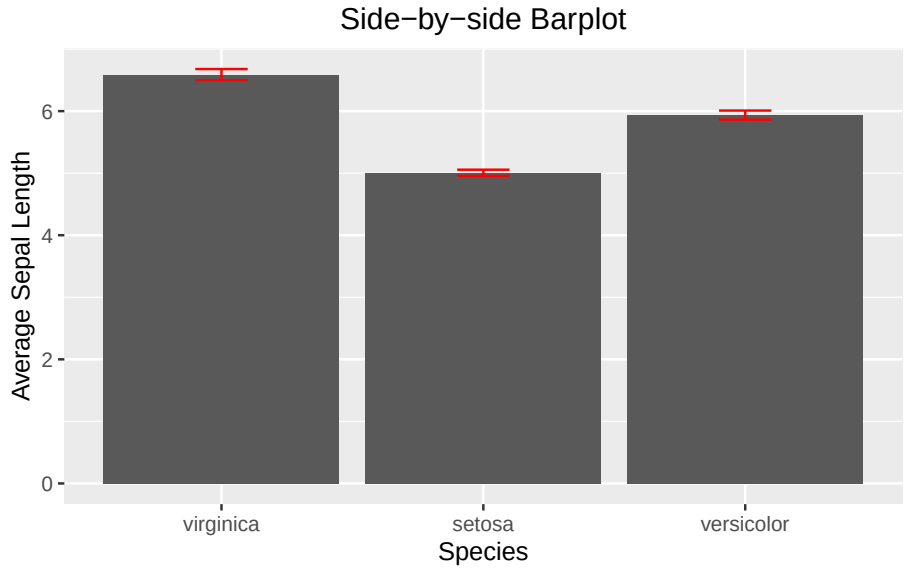



Figure 3.39: Side-by-side barplot with error bar

#### 3.3.4.4 Line Graph with Error Bar

```
iris_summary3 <- iris %>%
                 group_by(Species) %>%
                 summarize(mean_sepal_length = mean(Sepal.Length),
                           N=n(),
                           se=sd(Sepal.Length)/sqrt(N)) %>%
                 mutate(ymin=mean_sepal_length-se,
                        ymax=mean_sepal_length+se)
iris_summary3
```

```
# A tibble: 3 x 6
  Species    mean_sepal_length     N     se  ymin  ymax
  <ord>                  <dbl> <int>  <dbl> <dbl> <dbl>
1 virginica               6.59    50 0.0899  6.50  6.68
2 setosa                  5.01    50 0.0498  4.96  5.06
3 versicolor              5.94    50 0.0730  5.86  6.01
```

```
ggplot(iris_summary3, aes(x=Species, y=mean_sepal_length, group = 1)) +
  geom_line(color = 'skyblue') +
  geom_point(size = 3, color = 'skyblue') +
  geom_errorbar(aes(ymin = ymin, ymax = ymax), width = 0.1, linewidth = 1) +
  labs(x="Species", y="Sepal Length", title = "Line Plot with Erorr Bar") +
  theme(plot.title = element_text(hjust = 0.5))
```

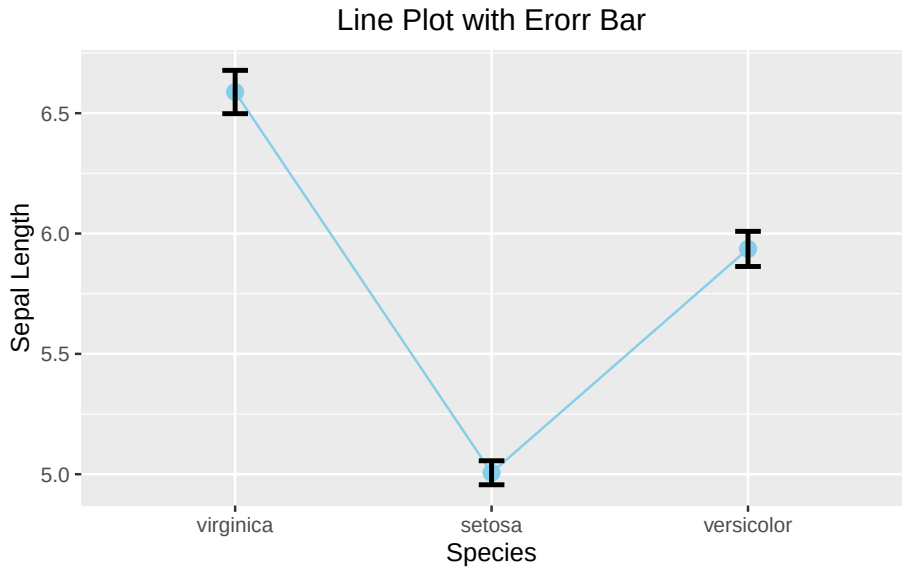



Figure 3.40: Line plot with error bar

#### 3.3.4.5 Overlapping Histogram

Like the base R side-by-side histogram, you will set the transparency to 0.5 and bin width to 0.25. Note that there is no need to care about the x- and y-limit in `ggplot2`.

```
ggplot(data=iris, aes(x=Sepal.Length, fill=Species)) +
  geom_histogram(alpha=0.5, binwidth = 0.25) +
```

```
  labs(x="Sepal Length", title = "Overlapping Histogram") +
  theme(plot.title = element_text(hjust = 0.5))
```

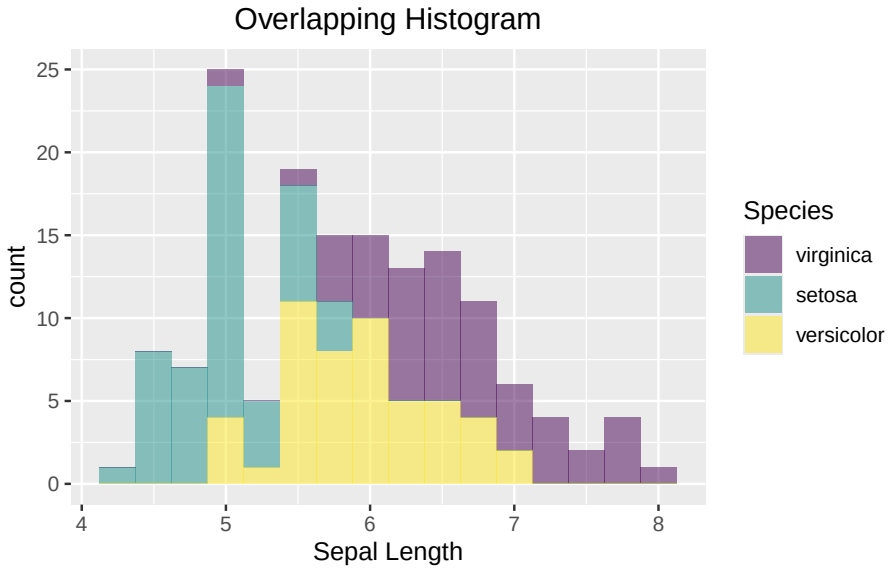



Figure 3.41: Side-by-side Histogram

### 3.3.5 Contingency Table

We will use the `penguins` data set to demonstrate how to create a side-by-side bar plot of the counts, and a mosaic plot.

First, let's create a contingency table as below:

```
mytab <- with(penguins, table(species, island))
mytab
```

```
           island
species     Biscoe Dream Torgersen
  Adelie        44    56        52
  Chinstrap      0    68         0
  Gentoo       124     0         0
```

As said above, `ggplot2` only works with data frames. Therefore, you need to convert the contingency table into a data frame with column names. During the conversion process, the R function will pair up each species with an island and

create a new column that stores the count. It means that the resultant data frame will consists of RxC rows, where R and C are the number of rows and columns, respectively. A 3x3 contingency table like yours, will be converted to a data frame consisting of 9 rows as shown below:

```
mydf <- as.data.frame(mytab)
colnames(mydf) <- c("species","island","Count")
mydf
```

```
    species    island Count
1    Adelie    Biscoe    44
2 Chinstrap    Biscoe     0
3    Gentoo    Biscoe   124
4    Adelie     Dream    56
5 Chinstrap     Dream    68
6    Gentoo     Dream     0
7    Adelie Torgersen    52
8 Chinstrap Torgersen     0
9    Gentoo Torgersen     0
```

```
ggplot(data=mydf, aes(x=island, y=Count, fill=species)) +
  geom_col(position = "dodge") +
  labs(title = "Side-by-side Barplot", x = "Island", y = "Count")
```

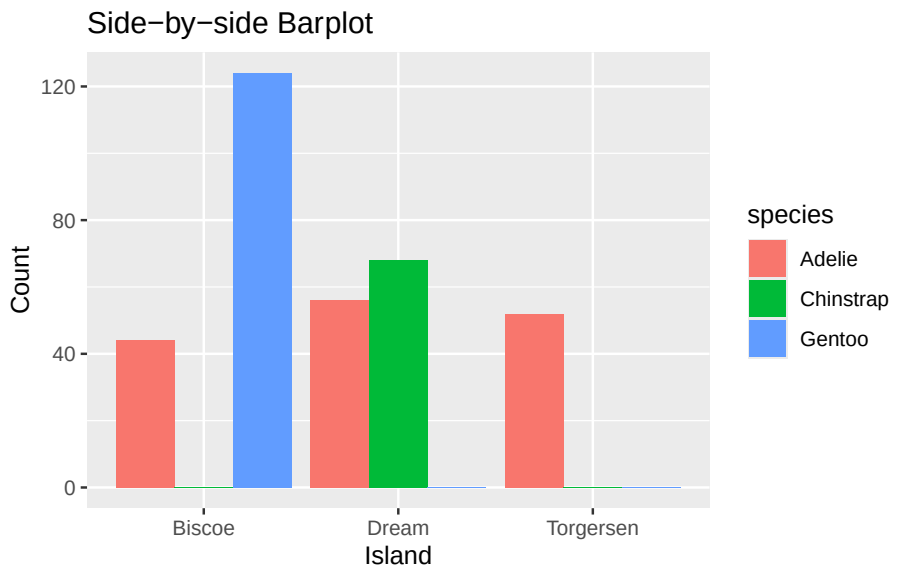



Figure 3.42: Side-by-side Barplot

### 3.3.6 Two Numeric Variables

Here, you are going to visualize the relationship between two numeric variables using a scatter plot. As said above, the revealed relationship can be linear and non-linear. Visualizing how data points are distributed on a scatter plot is the most intuitive way to understand their true relationship.

```
ggplot(iris, aes(x=Sepal.Width, y=Sepal.Length)) +
  geom_point() +
  labs(x="Sepal Width", y="Sepal Length", title = "Scatter Plot") +
  theme(plot.title = element_text(hjust = 0.5))
```

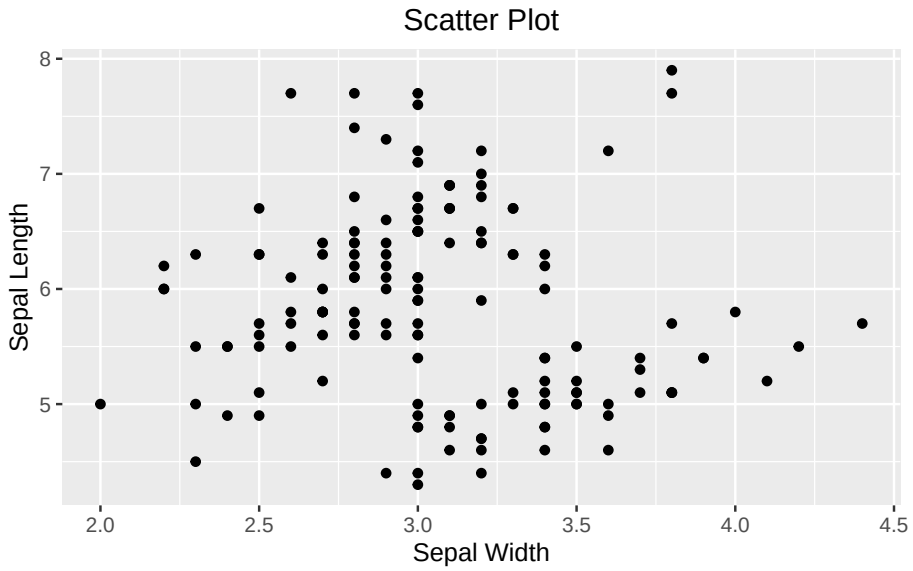



Figure 3.43: Visualizing the relationship between two numeric variables using a scatter plot.

If you wanted to emphasize their linear relationship, it can be done convincingly by adding a regression line on top of the scatter plot. `ggplot2` spares you the effort to create a linear model like above to determine what the slope and intercept are. Instead, it can be done by adding the `geom_smooth()`. You tell `geom_smooth(method = "lm", se = FALSE, color = "red")` to use a linear method (`lm`) for smoothing the data points. See the code below:

```
ggplot(iris, aes(x=Sepal.Width, y=Sepal.Length)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
```

```
  labs(x="Sepal Width", y="Sepal Length", title = "Scatter Plot with the Regression Line") +
  theme(plot.title = element_text(hjust = 0.5))
```

```
`geom_smooth()` using formula = 'y ~ x'
```

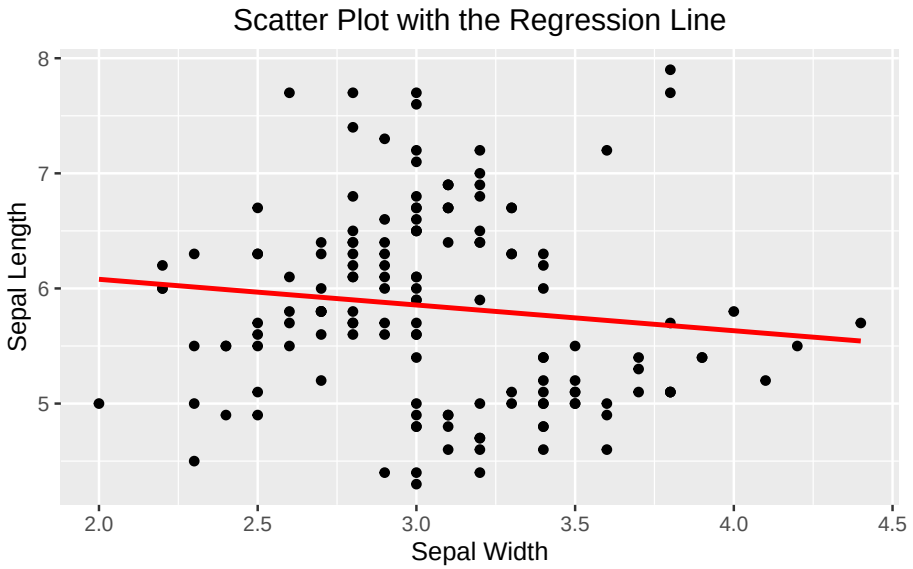



Figure 3.44: A scatter plot overlaid with a regression line.

If you see this message "`geom_smooth()` using formula = 'y ~ x' ", it is harmless, and informational. `ggplot2` is just to let you know what the response and predictor are used by `geom_smooth()` in creating the regression line.

### 3.3.7 Multi-panel Scatter Plot

Often times before multiple linear regression, it is a good idea to visualize the linear relationship between the response variable and the independent variables pairwisely. Or, the linear relationship between independent variables, as it may cause the collinearity issue in which independent variables are linearly correlated. A multi-panel scatter plot can do this. However, `ggplot2` does not have a geometry function for it. The simplest way is to use the `ggpairs()` function provided by the `GGally` package, which requires installation, if you have not done so.

Suppose, you are interested in the pairwise relationship between all numeric variables in `iris`. It can easily be done using the `ggpairs()` plot function.

Color is the attribute used to distinguish species. `progress = FALSE` is to suppress the printing of the progress bar.

```
library(GGally)
ggpairs(iris, aes(color = Species), title = "A Multi-panel Scatter Plot", progress = F
```

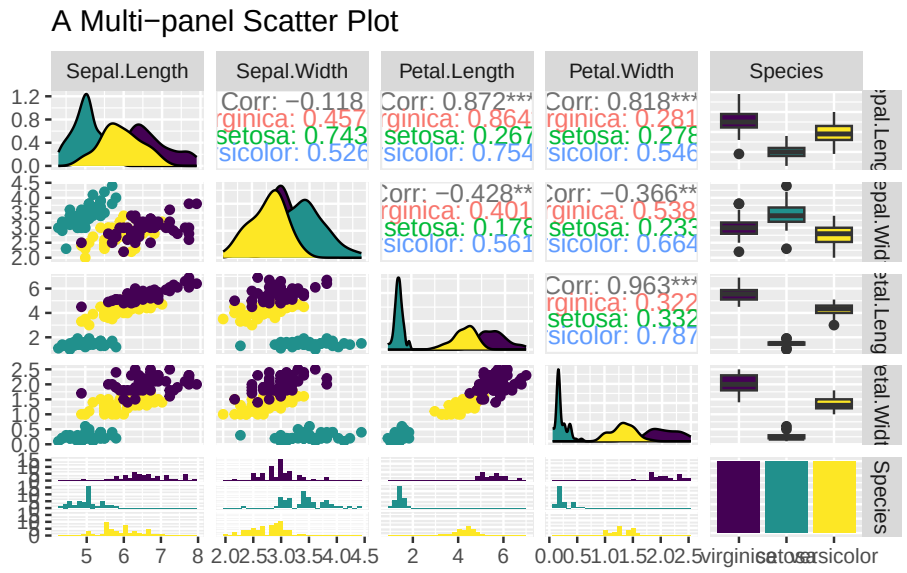



Figure 3.45: A Multi-panel Scatter Plot

## 3.4 Summary

This is big chapter that has discussed an array of plot functions both from base R and `ggplot2`. It is more like a cookbook of various visualization recipes. So, instead of listing all the functions here, it is helpful to list several useful resources that you can fine-tune the graphs yourselves.

**`pch`**

If you type `help(pch)` at the console, R shows the `pch` section of the help page for `points`. Below is a snapshot of `pch` values and the corresponding plotting characters:

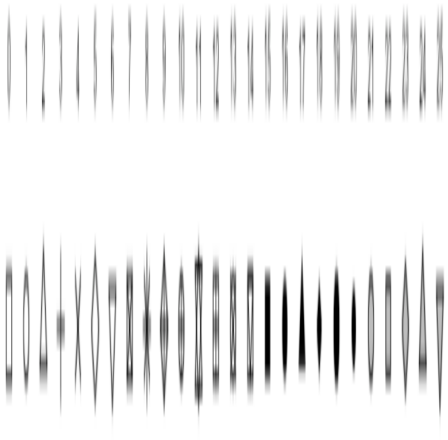


Figure 3.46: `pch` values

### Color

R supports a large number of colors and palettes. Here is a great resource to see what are available and their names: https://r-graph-gallery.com/colors.html.

# Chapter 4

# Dive Deeper

So far, we have discussed the basic functions of R that are sufficient for most situations. Here, we will discuss several intermediate level features that are practical, and yet utilize additional R functions. That includes the how to create summary tables from published work, converting a dataset from a wide format to a long format, and vice versa, the family of `apply()` functions, and a second look at `group_by()`.

## Learning Objectives:

- Recreate a summary table from published work in R.
- Create a small data frame with data in R.
- Load datasets into R.
- Convert a dataset from a wide format to a long format.
- Convert a dataset from a long format to a wide format.
- Utilize the family of `apply()` functions.
- Revisit `group_by()`.

## 4.1 Recreate a Summary Table in R

If you see a data summary table as one in Figure (@ref(fig:Fig4-1)), and you may want to reproduce the analysis. Unless the raw data, i.e., the data for each sample, is available, it cannot be reproduced in R using the `table()` or

`group_by()` functions. In such a scenario, you will learn the way to create an R table in this section.

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by agegp and tobgp.
i Output is grouped by agegp.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(agegp, tobgp))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 4 x 4
# Groups:   agegp [1]
  agegp tobgp    Controls Cases
  <ord> <ord>       <dbl> <dbl>
1 45-54 0-9g/day       90    14
2 45-54 10-19          44    13
3 45-54 20-29          25     8
4 45-54 30+             8    11
```

A

| Heart Disease | Smoking Status | | | |
|---|---|---|---|---|
| | Formerly | Never | Smoke | Unknown |
| 0 | 808 | 1802 | 728 | 1496 |
| 1 | 77 | 90 | 61 | 48 |

B

| Group | Daily Tobacco Consumption | Controls | Cases |
|---|---|---|---|
| A | 0-9g/day | 90 | 14 |
| B | 10-19 | 44 | 13 |
| C | 20-29 | 25 | 8 |
| D | 30+ | 8 | 11 |

Figure 4.1: Recreate summary tables.

Here you will create the two tables in R as shown in Figure (@ref(fig:Fig4-1)). Let us begin with the first table, which consists of four columns and two rows. As all the values are numeric, you can create a numeric vector for each row using the combine or collate function `c()` (@ref(vector). But in here, you will

specify the vectors a bit differently. To include the column names in the final table, you have to associate a name to each value for the first row. For the subsequent rows, make sure the order of the values corresponds to the intended columns. As a result, each column in the final table will have a name when rows are combined. Below is the R code to create two rows with names using a vector:

```
row1 <- c(heart_disease=0, formerly=808, never=1802, smoke=728, unknown=1496)
row2 <- c(1, 77, 90, 61, 48)
```

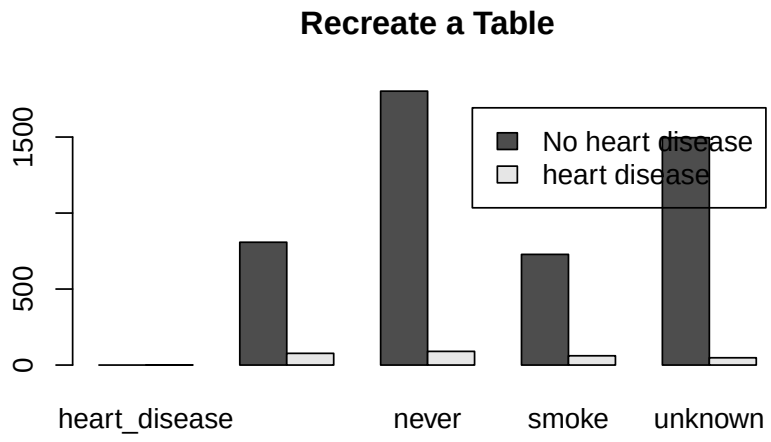
Next, you create a table by stacking up the rows using the `rbind()` function as shown below.

```
my_tab1 <- rbind(row1, row2)
my_tab1
```

```
     heart_disease formerly never smoke unknown
row1             0      808  1802   728    1496
row2             1       77    90    61      48
```

For example, you can plot the data, perform a $\chi^2$ test of independence, etc.

```
barplot(my_tab1, main="Recreate a Table", beside=T, legend.text = c("No heart disease","heart dis
```



```
chisq.test(my_tab1)
```

```
Warning in chisq.test(my_tab1): Chi-squared approximation may be incorrect
```

```

    Pearson's Chi-squared test

data:  my_tab1
X-squared = 61.967, df = 4, p-value = 1.119e-12
```

For the second table, each row contains a mixture of character and numeric values. Therefore, vector objects cannot be used to represent rows @ref(atomic-and-object-variables). Instead, use `list` objects for the **heterogeneous** rows. Below is an example to define rows as list objects:

```
A <- list(group="A", daily_toba="0-9", ncontrols=90, ncases=14)
B <- list("B", "10-19", 44, 13)
C <- list("C", "20-29", 25, 8)
D <- list("D", "30+", 8, 11)
```

Similarly, rows as lists are stacked up together to form a table.

```
my_tab2 <- rbind(A,B,C,D)
my_tab2
```

```
  group daily_toba ncontrols ncases
A "A"   "0-9"      90        14
B "B"   "10-19"    44        13
C "C"   "20-29"    25        8
D "D"   "30+"      8         11
```

## 4.2 Create a small data frame with data in R

In this section, you will learn two approaches to store a small dataset in an R's `data.frame` object. If you have a large dataset, this approach might not be the best choice. Instead, you should prepare the dataset using a spreadsheet software, e.g., Google Sheets or Microsoft Excel. Read the next section for more details.

### 4.2.1 Use `data.frame()` Function

Besides reproducing a table, you might want to create a small dataset as a data frame in R. It can be done in two steps, like creating a table above. The first step is to define each column (variable) as a vector, and then use the `data.frame()` function to combine them into a data frame. Here is an example to create a data frame that consists of two columns (group, and systolic blood pressure (sbp)), where each column contains eight samples. The first step is to create two vectors, one for each column

```
group_data <- c("A","A","A","A","A","B","B","B")
sbp_data <- c(120,135,133,128,130,150,160,158)
```

Next, use the `data.frame()` function to combine the vectors into a data frame.

```
small_df1 <- data.frame(group = group_data,
                        sbp = sbp_data)
small_df1
```

```
  group sbp
1     A 120
2     A 135
3     A 133
4     A 128
5     A 130
6     B 150
7     B 160
8     B 158
```

### 4.2.2 Use `read.table()` Function

An equivalent approach is to type out the values in columns as the example below, and feed it to the `read.table()` function through the `text=` parameter.

```
content = ("
group sbp
A 120
A 135
A 133
A 128
A 130
B 150
B 160
```

```
B 158
")

small_df2 <- read.table(text = content, header = TRUE)
small_df2
```

```
  group sbp
1     A 120
2     A 135
3     A 133
4     A 128
5     A 130
6     B 150
7     B 160
8     B 158
```

## 4.3 Load datasets into R

In this section, you will learn several ways to load a data file into R. Data files are usually in two formats: comma-separated (`.csv`) or tab-separated (`.txt` or `.tsv`). In a `.csv` file, comma (`,`) is used as the separator to separate data in one column from the other column as shown in Figure 4-2A (@ref(fig:Fig4-2)). The other commonly used separator is the `tab` symbol, which is usually invisible, but it is made visible deliberately in Figure 4-2B.

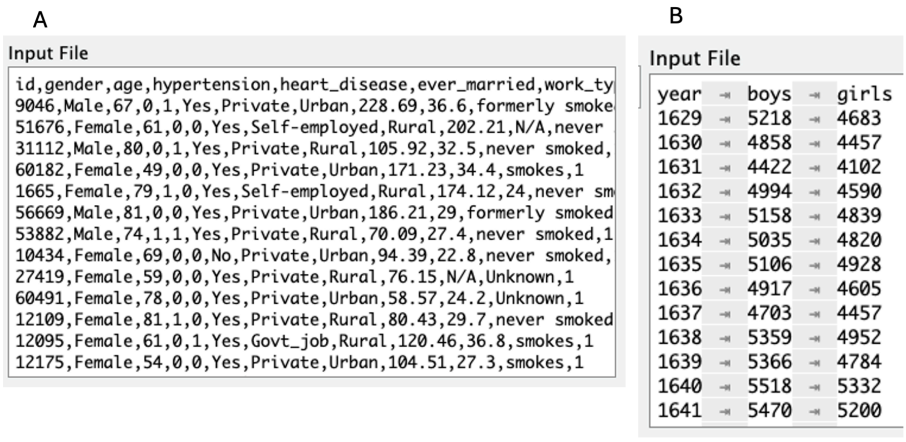



Figure 4.2: File formats.

### 4.3.1 `read.csv()` Function

### 4.3.2 Import data in RStudio

## 4.4 `group_by()`

## 4.5 Different `apply` Functions

## 4.6 Wide to Long

Use AirPassengers dataset

# Bibliography